

Attention and Regret^{*}

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Abstract

This paper develops an evolutionary foundation for regret theory. Our model considers agents who underinvest in attention (e.g., due to present bias) and studies which emotional responses to decision outcomes would best incentivize appropriate attention. We prove that regret emerges as the optimal emotion. Our evolutionary approach provides a criterion to arbitrate between different versions of regret theory and generates new predictions. Notably, regret exhibits a self-blame dimension: regret is stronger for decision problems where the right choice was easier to discover.

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[Regret] may also depend on the extent to which the individual blames himself for his original decision. [...] The neglect of this dimension of regret—although a useful simplifying assumption for many problems—is a serious obstacle to the development and generalisation of regret theory. - Sugden, 1986

I Introduction

Decisions under uncertainty can trigger regret and rejoicing when one learns that another action would have done better or worse. These emotions are often viewed as futile “crying over spilled milk,” and their anticipation has been shown to distort choices, predicting violations of expected utility theory (Bell, 1982; Loomes and Sugden, 1982; Zeelenberg, 1999a). Yet emotions research has increasingly emphasized that emotions serve important functions (Keltner and Gross, 1999), motivating our question: Why do regret and rejoicing exist if only actual outcomes should matter from an evolutionary perspective?

Although the anticipation of regret and rejoicing can distort decision-making, we show that these emotions can nevertheless improve the quality of decisions if we take into account *attention*. When facing uncertainty, individuals can improve decision-making by paying costly attention, which we understand broadly as encompassing both external information acquisition and internal information processing such as deliberation, planning, or memory search. The anticipation of regret and rejoicing can motivate individuals to pay more attention to avoid regret and seek rejoicing. Such an additional motive to pay attention is beneficial, if the agent is biased toward insufficient attention due to the temptation to avoid attention effort or due to present bias, as benefits are only realized after the cost of attention is incurred.

To investigate our hypothesis that regret evolved to motivate attention, we ask whether regret is indeed the optimal emotion for this purpose. This is not clear *ex ante* because any emotion that rewards good outcomes and penalizes bad outcomes would arguably raise the stakes of decisions and thereby motivate attention. For instance, this would also be achieved by disappointment and elation, which also depend on counterfactual comparisons, but to other states of the world rather than other actions. Using a mechanism-design approach, we show that regret and rejoicing emerge as optimal from a non-parametric class of emotions and that their distinctive structure—counterfactual comparisons to alternative actions—is precisely what makes them optimal for motivating attention. We then leverage our functional approach to deliver a precise theory of regret. This theory connects closely to existing regret theories while enriching them by explicitly accounting for attention.

We establish our optimality result in a model that follows the principal-agent approach to the evolution of preferences.¹ In this approach, evolution—whether natural or cultural—is modeled as a principal who “designs” the agent’s preferences to induce fitness-maximizing behavior. In our model, the principal delegates decision-making to an agent who overweights attention costs relative to decision outcomes. To incentivize attention, the principal endows the agent with emotional utility

¹The principal-agent approach to the evolution of preferences has been employed, among others, by Robson (2001), Samuelson and Swinkels (2006), Rayo and Becker (2007), Netzer (2009), and Baliga and Ely (2011).

as a function of the chosen action and realized state, which enters the agent’s utility in addition to material utility. Crucially, emotional utility cannot condition on attention itself, inducing a hidden-action problem, nor on the agent’s bias, inducing a hidden-information problem. These constraints reflect that attention is harder to verify than outcomes and that the strength of the bias may be uncertain due to stochastic temptation or heterogeneity in the population. We model attention using the rational-inattention paradigm, where the agent can choose any signal structure subject to an entropy-based information cost (Sims, 2003). This creates a high-dimensional mechanism-design problem with both hidden information and hidden actions. To render it tractable, we restrict ourselves to decision problems that are invariant under relabeling actions, and impose that the emotional utility preserves this symmetry. We discuss our model assumptions in detail in Section II.

Regret and rejoicing are optimal because they amplify utility differences between actions depending on the material stakes in each state. Formally, let $v(a, \omega)$ denote the material payoff under action a and state ω . In our model, regret or rejoicing experienced in state ω under action a is a function of the vector of payoff differences $v(a, \omega) - v(a', \omega)$ to alternative actions a' under the realized state ω . By rewarding better and penalizing worse actions, regret and rejoicing amplify the state-wise utility differences between actions, which we refer to as the **stakes** in that state. The emotional amplification of stakes is optimal because it motivates more attention to the state. Crucially, regret and rejoicing amplify these stakes by an amount that depends on the material payoff differences between actions, that is, the material stakes. By depending on material stakes, these emotions optimally direct attention to states where stakes are higher.

For binary decision problems, the canonical setting in regret theory, we obtain sharper results. The optimal emotions yield *original regret theory* (Bell, 1982; Loomes and Sugden, 1982), including its key assumption, disproportionate aversion to large regrets. This assumption means that regret amplifies larger utility differences between actions disproportionately. Disproportionate aversion to large regrets explains well-documented violations of expected utility theory, including the Allais paradox, the common ratio effect, and simultaneous gambling and insurance (Bell, 1982; Loomes and Sugden, 1982). Our model predicts a weaker version of this property—starshapedness rather than convexity—but we demonstrate that starshapedness remains sufficient to generate the above violations of expected utility that motivated the convexity assumption.

In our model, disproportionate aversion to large regrets arises from heterogeneity in the agent’s bias toward insufficient attention. When the material stakes are already large, agents with small biases choose the ex-post optimal action most of the time, and their mistake rates are relatively insensitive to further increases in the stakes. Agents with stronger biases make more frequent mistakes, and their behavior is more responsive to stake increases. The principal therefore tailors the emotional amplification to the more biased types by scaling up larger material stakes disproportionately more.

Our model addresses two important limitations of original regret theory.

First, our model captures the self-blame aspect of regret, which original regret theory inade-

quately addresses, as argued by one of its pioneers (Sugden, 1986) and emphasized by decision-justification theory (Connolly and Zeelenberg, 2002). Taking self-blame seriously, regret theory should account for whether one could have known that another action would turn out better. Our model includes a complexity parameter for the decision problem that captures how costly it is to discover the right action. The model indeed predicts that when it is less costly to discover the right action, mistakes are punished more severely by regret.

Second, our model predicts regret beyond binary choice, which has proven difficult for original regret theory (Loomes and Sugden, 1982; Loomes and Sugden, 1987). A frequently used model for general choice sets considers only regret relative to the ex-post best action (Quiggin, 1994). Our model makes the psychologically plausible prediction that regret and rejoicing are affected by all counterfactual actions, in line with generalized regret theory (Loomes and Sugden, 1987; Sugden, 1993). Our model strengthens generalized regret theory by predicting that regret depends only on material payoff differences. Further, it predicts that regret maintains the utility ordering of actions.

Our evolutionary foundation for regret has implications for how to evaluate regret and contributes to a broader debate on whether emotions are adaptive (Keltner and Gross, 1999). Knowing the benefits and costs of regret is useful, when we can influence these emotions through socialization or therapy, or through providing counterfactual feedback. Psychological studies frequently theorize that regret is an adaptive emotion to improve decision-making (e.g., Janis and Mann, 1977; Zeelenberg, 1999b; O’Connor, McCormack, and Feeney, 2012). This hypothesis is supported empirically by Camille et al. (2004), who find that patients who do not experience regret as the result of a damage to their orbitofrontal cortex make worse decisions. We contribute by providing an attention-based mechanism through which regret improves decisions and by showing that this mechanism can even provide an optimality foundation for regret. Our attention-based mechanism is empirically supported by Reb (2008), who finds that making regret more salient through providing counterfactual feedback leads subjects to take longer to reach a decision and to acquire more information.

A popular alternative hypothesis for how regret may benefit decision-making is that it improves learning in repeated decisions. When agents experience regret, they purportedly learn to modify their future behavior. This hypothesis draws theoretical support from reinforcement learning with regret, called regret matching, which has desirable asymptotic properties (Foster and Vohra, 1999).² However, we are not aware of any optimality foundation for regret matching in this framework. Furthermore, the connection between regret in reinforcement learning and regret as conceptualized in decision theory remains tenuous, both in terms of behavior and the way regret is represented. Regret matching predicts stochastic behavior where the probability of an action is proportional to its historical regret. By contrast, regret theory predicts that agents choose the action with the highest expected utility, where utility has an emotional component. Our framework provides an optimality foundation and has a strong link to regret theory, in particular, its original version due

²Regret also appears prominently in reinforcement learning as a performance benchmark for learning algorithms (“no regret learning”). Regret matching is one of many learning algorithms that achieves this benchmark.

to Bell (1982) and Loomes and Sugden (1982).

Related Literature Regret theory was introduced by Bell (1982) and Loomes and Sugden (1982). Their original model, developed for pairwise choices, relies on a convexity assumption on regret and accounts for several empirical violations of expected utility theory. Bleichrodt, Cillo, and Diecidue (2010) show that regret can be measured using the trade-off method and confirm the convexity assumption experimentally. Building on the trade-off method, Diecidue and Somasundaram (2017) provide an axiomatization of original regret theory, strengthening earlier axiomatizations for more general regret models (Fishburn, 1989; Sugden, 1993). Bleichrodt and Wakker (2015) review further empirical support and applications of regret theory across domains such as insurance, health, finance, and auctions. This paper contributes to this literature by providing a foundation for original regret theory (subject to one modification) and by showing how this foundation can be used to inform the modeling of regret.

At a broader level, our paper contributes to the literature on how a principal should incentivize an expert to acquire information, beginning with Demski and Sappington (1987). Our model allows for flexible information acquisition, as in recent contributions such as Lindbeck and Weibull (2020) and Yoder (2022), and is distinctive in combining this with both hidden actions (signal structures are not contractible) and hidden information (the agent’s bias is private). We contribute to this literature by showing that counterfactual comparisons (as they factor into regret) are useful for incentivizing information acquisition.

Yoder (2022) also considers flexible information acquisition and hidden information, but assumes that either the signal structure or the signal realization is contractible. In contrast, in our model only the action and realized state are contractible. We show that self-screening is ineffective and hence our solution methods and the resulting distortions differ: in his model, there is no distortion at the top, whereas in our case, information acquisition is skewed towards extreme states for all types. Methodologically, Yoder (2022) uses the posterior approach (Caplin, Dean, & Leahy, 2022). While we use the posterior approach to show that self-screening is ineffective in our setting, for the bulk of our results we use the state-dependent choice probabilities approach (Matějka & McKay, 2015) and its connection to perturbed utility models (Fosgerau, Melo, de Palma, & Shum, 2020).

The paper is organized as follows. Section II introduces and discusses the model. Section III presents the results under two and more than two actions. Section IV concludes.

The Appendix provides proofs and extensions. Appendix A contains proofs. Appendix B takes initial steps to extend the model to incomplete ex-post feedback about the state of the world, where disappointment and elation arise to supplement regret and rejoicing to motivate attention. Appendix C discusses how our model relates to more standard principal-expert problems in economics.

II Model

Decision Problem The principal (evolution, she) faces a *decision problem*, which is a tuple (Ω, μ, A, v) . It comprises a finite state space Ω , a full-support prior $\mu \in \Delta(\Omega)$ on the state space, a finite set of actions A , and a utility function $v: A \times \Omega \rightarrow \mathbb{R}$, which we refer to as the material payoff.³

Attention The principal delegates a two-stage problem to the agent (he). First, he acquires a signal structure about the state (the attention stage) and, second, he chooses an action from A conditional on the realized signal (the decision-making stage). A signal structure is a pair (S, σ) , where S is a finite signal space and $\sigma: \Omega \rightarrow \Delta(S)$ is a state-dependent distribution over signals, with $\sigma(s|\omega)$ denoting the probability of signal s in state ω . Following the rational-inattention approach (Sims, 2003), we assume that the agent can select *any* signal structure subject to the *mutual information* cost, defined as

$$c(S, \sigma) := \sum_{s \in S} \left(-\sigma(s) \log \sigma(s) \right) - \sum_{\omega \in \Omega} \left(\mu(\omega) \sum_{s \in S} \left(-\sigma(s|\omega) \log \sigma(s|\omega) \right) \right).$$

Here, $\sigma(s) := \sum_{\omega \in \Omega} \mu(\omega) \sigma(s|\omega)$ is the unconditional probability of signal $s \in S$ and $\log$ refers to the natural logarithm. As usual, we define $0 \log 0 = 0$. The first sum is the entropy of the unconditional signal distribution, while the second is the expected entropy of the conditional signal distribution given the state. Intuitively, entropy quantifies the uncertainty of a distribution and mutual information measures how much knowing the state reduces the uncertainty about the signal. Hence, the attention cost reflects the extent to which the signal is tailored to the state.

It is well known (e.g., Matějka and McKay, 2015) that the agent's signal can be identified with an action recommendation. The reason is that the mutual information cost is Blackwell monotonic, so the agent profits from garbling a signal structure based on its action recommendation. This allows one to reduce the agent's two-stage problem to the one-stage problem of selecting state-dependent choice probabilities $p: \Omega \rightarrow \Delta(A)$, written $p(a|\omega)$. The state-dependent choice probabilities p , which we also refer to as an *attention strategy*, entail an attention cost $c(p)$ that corresponds to the mutual information of the action and state,

$$c(p) := \sum_{a \in A} \left(-p(a) \log p(a) \right) - \sum_{\omega \in \Omega} \left(\mu(\omega) \sum_{a \in A} \left(-p(a|\omega) \log p(a|\omega) \right) \right).$$

Here, $p(a) := \sum_{\omega \in \Omega} \mu(\omega) p(a|\omega)$ is the unconditional probability of a . In sum, the attention cost $c(p)$ captures how much the action is tailored to the state.

³We assume a finite state space to circumvent measure theoretical issues. We believe that our results carry over to infinite state spaces under the appropriate technical assumptions.

Preferences The principal's utility V derived from an attention strategy p is

$$V(p) = \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p(a|\omega) v(a, \omega) - \lambda c(p).$$

That is, she internalizes the attention cost, weighted by λ , and derives utility $v(a, \omega)$ from action a under state ω . We interpret λ as the difficulty of the decision problem, as it scales the cost of learning about the state. This parameter will be used for performing comparative statics.

In our model, from the principal's perspective, the agent has an inherent bias, represented as a discount factor $\beta \in (0, 1)$ applied to the material payoff $v(a, \omega)$. Equivalently, this bias can be viewed as overweighting the attention cost by a factor of $1/\beta > 1$. In our main interpretation, the bias captures a present bias or the temptation to avoid the attention cost. Under the alternative cultural-evolutionary interpretation discussed below, the bias may also reflect conflicts between parents and children due to paternalistic altruism.

We depart from standard delegation models in two respects. First, the bias β is uncertain and private information of the agent, with cumulative distribution function F . This uncertainty can be due to incomplete knowledge of the agent's fixed bias as in models of self-signaling (Bénabou and Tirole, 2004) or due to the stochastic nature of the bias, as in random-indulgence models (Chatterjee and Krishna, 2009; Duflo, Kremer, and Robinson, 2011; Dekel and Lipman, 2012). Second, the principal can use action- and state-contingent payoff-transfers $m(a, \omega)$ to incentivize the agent. We interpret these transfers as *emotions*, which can depend on the chosen action and the realized state.⁴ The agent with bias β derives utility U_β from the attention strategy p ,

$$U_\beta(p) = \beta \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p(a|\omega) (v(a, \omega) + m(a, \omega)) - \lambda c(p).$$

Equivalently, the agent maximizes

$$\frac{U_\beta(p)}{\beta} = \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p(a|\omega) (v(a, \omega) + m(a, \omega)) - \frac{\lambda c(p)}{\beta}.$$

We make three assumptions on the distribution of the bias. First, the distribution of the agent's type β is non-degenerate, so there is some uncertainty about the bias of the agent. Second, the bias is bounded away from zero, which ensures that finite emotions are optimal. Third, the distribution of β is discrete, which simplifies some proofs.

Assumption 1. *The bias β has a non-degenerate, discrete distribution with support $\text{supp}(\beta) \subseteq [\underline{\beta}, 1]$ with $\underline{\beta} > 0$.*

⁴As common in many models, we assume that emotions are additively separable in utility (e.g. Bell, 1982, 1985; Loomes and Sugden, 1982, 1986). We also make this assumption because it allows the principal to implement any emotion-inclusive utility function $u(a, \omega) = v(a, \omega) + m(a, \omega)$. Up to the determination of a state-wise constant, this assumption is immaterial for the model's prediction of u .

Tie-Breaker Thus far, the model can only identify optimal emotions up to a state-wise constant. Given that the agent is an expected-utility maximizer and the attention cost is additively separable, shifting all emotional utilities in a given state by a constant does not alter his behavior.⁵ Moreover, emotions do not enter the principal’s utility directly because, as usual, evolution just cares about the induced behavior. Thus, the principal is indifferent between any shifts of emotions by state-wise constants. While such constants do not affect the agent’s behavior, they do matter for welfare comparisons. To resolve the principal’s indifferences, we introduce a tie-breaker that determines to what degree the principal uses positive or negative emotions to motivate the agent.

Our tie-breaker is based on an implementation cost $\psi: \mathbb{R} \rightarrow \mathbb{R}$ for positive as well as negative emotions. From the possible solutions to the principal’s problem, we refine the principal’s choice to the mechanism that minimizes the expected cost of emotion $\Psi(m)$,

$$\Psi(m) := \int \left(\sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p_{\beta}(a|\omega) \psi(m(a, \omega)) \right) dF(\beta),$$

where ψ is strictly convex, differentiable and symmetric around 0.^{6,7} The cost of emotions $\psi(m)$ could be due to the energy used to induce emotions in the brain or due to the opportunity cost from diminished capacity for other cognitive processes. Under the cultural-evolution interpretation, the cost can be interpreted as the cost of inculcating emotions through parenting and socialization, similar to Kaplow and Shavell (2007).

Assumption 2. *The cost of emotions $\psi: \mathbb{R} \rightarrow \mathbb{R}$ is strictly convex, differentiable, and symmetric around 0.*

Note that under strict convexity, symmetry around 0 implies that the cost ψ is minimal at 0. Except for Corollary 2, which shows that regret is stronger than rejoicing, all our results hold assuming the cost is minimal at 0 without assuming symmetry. To ensure that the solution is shaped by the optimal incentives for attention rather than the implementation cost, we treat the minimization of the cost of emotions as a tie-breaker, instead of directly incorporating the cost of emotions into the principal’s utility function.

Exchangeability Due to the presence of both hidden actions and hidden information in the principal’s problem, the problem is very complex in general. In particular, the agent’s choice of an attention strategy is a high-dimensional problem and generally does not have a closed-form solution. To render the problem tractable, we make two symmetry assumptions. First, we impose that relative to the choice set A and prior μ , the utility v is *exchangeable* (Matějka and McKay,

⁵As in regret theory, *given emotions*, our agent is an expected-utility maximizer. However, because emotions can be a function of the choice set, the agent’s choice behavior can violate expected-utility axioms.

⁶Instead of a cost of emotions, one could also assume the budget constraint that emotions are 0 in expectation, for example due to hedonic adaptation. This budget constraint is in fact implied by assuming a quadratic cost of emotions $\psi(m) = m^2$.

⁷If one assumed instead that the principal internalizes the agent’s emotions positively, equivalent to setting $\psi(m) = -m$, then there would be no solution because the principal would prefer infinitely positive emotions.

2015). Second, we restrict ourselves to mechanisms where emotions maintain exchangeability of the utility. Exchangeable utility captures that the decision problem is invariant under relabeling the actions. Exchangeability helps characterize the agent’s attention strategy, which is given by a multinomial logit formula (Matějka and McKay, 2015).

Definition 1. A utility function $u: A \times \Omega \rightarrow \mathbb{R}$ is *exchangeable* if the distribution over payoff vectors $(u(a, \omega))_{a \in A} \in \mathbb{R}^A$ induced by prior μ is invariant under permutation of the coordinates.

As an example, the utility function according to the payoff-matrix in Table 1 is exchangeable if $\mu(\omega_1) = \mu(\omega_2)$ and $\mu(\omega_3) = \mu(\omega_4)$. That is because the payoff vector $(0, 1)$ is as likely as its permutation $(1, 0)$ and the payoff vector $(3, 2)$ is as likely as $(2, 3)$.

	ω_1	ω_2	ω_3	ω_4
a_1	1	0	3	2
a_2	0	1	2	3

Table 1: Exemplary payoff-matrix

Assumption 3. The material payoff v is exchangeable.

This symmetry assumption is natural, for example, when before any attention the actions are indistinguishable and this assumption should arguably hold at a sufficiently ex-ante stage. Under our main interpretation, this assumption is plausible if emotions are adapted to the general attention problem that the agent faces and not to specific contexts.

For tractability, we focus on emotions m that maintain exchangeability of the decision problem. Thus, the emotions do not depend on the labeling of actions. This symmetry assumption is natural because the underlying decision problem is symmetric in the actions, so the first-best attention strategy is symmetric in the actions.⁸

We thus study the following principal’s problem (P).

$$\max_{m, (p_\beta)_\beta} \int \left(\sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p_\beta(a|\omega) v(a, \omega) - c(p_\beta) \right) dF(\beta) \quad (\text{P})$$

s.t.

$$\forall \beta: p_\beta \in \arg \max_p \beta \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p(a|\omega) (v(a, \omega) + m(a, \omega)) - c(p) \quad (\text{IC})$$

$$v + m \text{ is exchangeable.} \quad (\text{EX})$$

Out of the solutions to (P), the principal chooses the one that minimizes the cost of emotions $\Psi(m)$.

One may wonder whether the principal could profit from offering a menu of emotions. This could allow the principal to tailor incentives to the agent’s type. Even more generally, the principal could elicit a message from the agent before and after attention prior to assigning an emotion, to

⁸Simulations under two states suggest that inducing asymmetries through emotions cannot benefit the principal. For a quadratic posterior-separable cost under two states, one can show that the asymmetric emotions are suboptimal.

screen the agent. However, we show, assuming that every emotion has to induce an exchangeable utility and under two states, that the principal *cannot* do better than offering a single emotion scheme, so the formulation of (P) above is without loss.⁹

Lemma 1. *Let $|\Omega| = 2$. Under Assumptions 1 to 3, any mechanism is weakly dominated by a single emotion scheme and the formulation (P) is without loss.*

All proofs are relegated to the Appendix. The proof of Lemma 1 shows that offering a menu makes types sort into emotion schemes to the principal’s disadvantage. Ideally, the principal would want to offer stronger incentives to types with lower β than to types with higher β because the former group is more biased. However, higher types have a higher return to attention and pick stronger incentives than lower types, contrary to the principal’s aim. This leads to higher types paying more attention and lower types less than under a single emotion scheme. Hence, offering a menu increases the variance of attention, decreasing the principal’s utility.

Interpretation and Discussion Our model follows the principal-agent approach to the evolution of preferences (e.g. Robson, 2001; Samuelson and Swinkels, 2006; Rayo and Becker, 2007). This approach likens evolution to a principal that chooses preferences to incentivize fitness-maximizing behavior while operating under certain constraints. Without constraints, evolution would simply hardwire optimal behavior and there would be typically be no meaningful results regarding preferences. In our model, these constraints are: (1) the agent’s bias cannot be directly mitigated, and (2) the degree of this bias and attention are private information of the agent. We discuss the justification for these constraints in the following.

Regarding the first constraint, we assume that evolution cannot directly mitigate the bias. Instead, emotions function as a kludge to improve decision-making. This assumption aligns with evidence for an incremental evolution of the mind, which adds new structures while leaving old structures intact (Linden, 2007; Marcus, 2008). Furthermore, this view is consistent with standard views of self-control issues as manifestations of a divided self. Dual-self models, also known as planner-doer models or dual-system models of the brain, conceptualize the organism as a far-sighted self interacting with a short-sighted self (Thaler and Shefrin, 1981; Bénabou and Pycia, 2002; Bernheim and Rangel, 2004; Fudenberg and Levine, 2006, 2012; Brocas and Carrillo, 2008). These two selves can be identified with evolutionary newer and older brain structures, such as the prefrontal cortex and the amygdala, which aligns with neuroscientific evidence of the prefrontal cortex exerting control over other brain structures (Miller and Cohen, 2001; see also Brocas and Carrillo, 2008).¹⁰ Compatible with this, neuroscientific research points toward a pivotal role of the prefrontal cortex in experiencing regret (Camille et al., 2004; Coricelli et al., 2005).

⁹We conjecture that this Proposition generalizes to any (finite) state space.

¹⁰While our main interpretation of the principal is evolution creating brain structures that induce regret, a dual-self interpretation is also possible. Novel compared to existing dual-self models is that our principal can use emotions as rewards or penalties to incentivize the agent. An exception is papers on goal-setting (Koch and Nafziger, 2011; Hsiaw, 2013), however, they constrain the choice of a utility function to specific functional forms.

Moreover, the coexistence of regret and the bias toward insufficient attention may persist as a *second-best equilibrium*, even if evolution could eliminate the bias. Once regret is in place, removing the bias would lead to excessive attention. Evolution can get stuck in an equilibrium where regret responds optimally to the bias and the bias responds optimally to regret. Waldman (1994) formally shows that such a second-best equilibrium can be evolutionarily stable, even if both traits—regret and the bias—can evolve at the same time. The reason is that even if a mutant possessed no regret nor bias, their offspring would have suboptimal trait combinations due to genetic recombination with the incumbent population’s traits.

Regarding the second constraint, we assume that evolution cannot condition emotions on the acquired signal structure or the stochastic bias β . It seems plausible that once the decision outcome materializes, actions are easier to recall and verify than pre-decision variables like temptation intensity or attention paid, as assumed in self-signaling models (Bodner and Prelec, 2003; Bénabou and Tirole, 2004). These non-contractibility constraints are crucial for our results. If the signal structure were contractible, the principal could achieve the first best by sufficiently penalizing any deviation from the optimal signal structure. If the bias were contractible, the principal could achieve the first best by scaling up payoffs by $1/\beta$. With neither contractible, regret emerges as the optimal emotion to motivate attention.

An alternative interpretation views the principal as parents, following the literature on intergenerational preference transmission (Bisin and Verdier, 2011, Doepke, Sorrenti, and Zilibotti, 2019). In this interpretation, the bias need not reflect a behavioral bias, but may refer to a conflict of interest between parents and children. Such conflicts are central to the literature on intergenerational preference transmission and arise from paternalistic preferences (Bisin and Verdier, 2001, Doepke and Zilibotti, 2017). A particularly salient conflict between parents and children concerns patience and future-oriented investments, “as witnessed by the relentless struggle of many parents to push their reluctant children to study diligently for school” (Doepke and Zilibotti, 2017). In our context, this manifests as parents caring more about future decision outcomes relative to attention costs than children do. If children place heterogeneous weights on attention costs relative to decision outcomes, parents face a random “bias.”¹¹ It is natural to suppose that parents observe only decision outcomes, not the child’s exact preferences or the amount of attention paid, since an important component of attention is internal information processing. Parents can then shape only preferences over decision outcomes, not the relative weight on decision outcomes versus attention costs. Preferences over decision outcomes may be shaped through blame or praise, which become internalized as self-blame (regret) or self-congratulation (rejoicing). As usual in economics, parents need not consciously solve our principal’s problem; successful parenting strategies can emerge through adaptive procedures like cultural evolution or learning.

Finally, our model can be applied to contractual relationships, where information acquisition is delegated to an expert and payoff-transfers are monetary instead of emotions. We discuss this

¹¹From an evolutionary perspective, heterogeneity of preferences may result from genetic variation (e.g., due to mutations), heterogeneous environmental influences, or as an evolutionary response to aggregate uncertainty (Cooper & Kaplan, 1982).

interpretation further in section VII.

III Results

First, we characterize the solution of (P) for decision problems containing two actions and then turn to the case of $n > 2$ actions.

A Two Actions

A.1 Regret Theory

To check whether the optimal emotions can be interpreted as regret and rejoicing, we introduce the two standard theories of regret for two actions, namely *original regret theory* due to Bell (1982) and Loomes and Sugden (1982), and *generalized regret theory* due to Loomes and Sugden (1987). This section closely follows Herweg and Müller (2021).

We start with **original regret theory**. The idea behind regret theory is that, when some state realizes, agents experience not only the realized material payoff but also an emotional utility depending on the material payoff difference to the counterfactual action. When choosing between two actions, agents compare the expectation of the emotions-inclusive utility over all states. Formally, original regret theory assumes a material payoff function $v : A \times \Omega \rightarrow \mathbb{R}$ and a regret-rejoicing function $R : \mathbb{R} \rightarrow \mathbb{R}$. Action a_i is weakly preferred to action a_j if and only if

$$\sum_{\omega \in \Omega} \mu(\omega) (v(a_i, \omega) + R(v(a_i, \omega) - v(a_j, \omega))) \geq \sum_{\omega \in \Omega} \mu(\omega) (v(a_j, \omega) + R(v(a_j, \omega) - v(a_i, \omega))) . \quad (1)$$

Here, $R(x)$ is interpreted as the emotional utility, which depends on the material payoff difference to the counterfactual action. The emotional utility is interpreted as regret if x is negative and rejoicing if x is positive. Given this interpretation, it is typically assumed that R is strictly increasing and $R(0) = 0$.

It is useful to rewrite (1) by subtracting the right-hand side from the left-hand side, which delivers

$$\sum_{\omega \in \Omega} \mu(\omega) Q(v(a_i, \omega) - v(a_j, \omega)) \geq 0, \quad (2)$$

where $Q(x) := x + R(x) - R(-x)$. This shows that the function Q captures all behavioral consequences of regret. Note that Q is by construction skew-symmetric, that is, $Q(-x) = -Q(x)$. Bell (1982) and Loomes and Sugden (1982) show that if Q is convex, then regret theory explains empirical violations from expected utility theory, such as the Allais paradox, the common-ratio effect, and simultaneous gambling and insurance. Original regret theory is axiomatized by Diecidue and Somasundaram (2017). Bleichrodt, Cillo, and Diecidue (2010) show how to experimentally identify v and Q from choices and empirically confirm the convexity assumption.

Definition 2. *The preferences between actions have an original regret representation if there is a*

material payoff function $v : A \times \Omega \rightarrow \mathbb{R}$ and a function $Q : \mathbb{R} \rightarrow \mathbb{R}$ that is skew-symmetric, strictly increasing, and strictly convex on $[0, \infty)$, such that for any two actions, a_i and a_j , a_i is weakly preferred to a_j if and only if (2) holds.

Generalized regret theory is introduced by Loomes and Sugden (1987) as a more general theory that circumvents the need for separate v and Q functions. To define it, we need additional notation for an ordered outcome $x(a, \omega)$ of action a in state ω , $x : A \times \Omega \rightarrow \mathbb{R}$. Usually, x refers to monetary outcomes. Generalized regret theory assumes a skew-symmetric function $\Psi : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$, that is, $\Psi(x, y) = -\Psi(y, x)$. Action a_i is weakly preferred to a_j if

$$\sum_{\omega \in \Omega} \mu(\omega) \Psi(x(a_i, \omega), x(a_j, \omega)) \geq 0. \quad (3)$$

Loomes and Sugden (1987) impose three assumptions on Ψ :

- **Ordering**: for all $x, y \in \mathbb{R}$: $x \geq y \Leftrightarrow \Psi(x, y) \geq 0$.
- **Increasingness**: for all $x, y, z \in \mathbb{R}$: $x \gtrless y \Leftrightarrow \Psi(x, z) \gtrless \Psi(y, z)$.
- **Convexity**: for all $x, y, z \in \mathbb{R}$: $x > y > z \Rightarrow \Psi(x, z) > \Psi(x, y) + \Psi(y, z)$.

Definition 3. *The preferences between actions have a generalized regret representation if there is a function $x : A \times \Omega \rightarrow \mathbb{R}$ and a function $\Psi : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ that is skew-symmetric and satisfies ordering, increasingness, and convexity, such that for any two actions, a_i and a_j , a_i is weakly preferred to a_j if and only if (3) holds.*

Fishburn (1989) and Sugden (1993) axiomatize generalized regret theory. Loomes and Sugden (1987) show that if Ψ satisfies Ordering, Increasingness, and Convexity, then generalized regret theory shares many of the predictions of original regret theory.

A.2 Optimal Emotion

This section considers binary action sets, $A = \{a_1, a_2\}$. Based on original regret theory, we define a regret solution to the principal's problem. However, we weaken the assumption that Q is convex to Q being **starshaped**. We comment on this difference below.

Definition 4. *A function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is **starshaped** if $\phi(0) = 0$, $\phi(x) > 0$ for $x > 0$, and if for all $x > 0$ and $k > 1$:*

$$\phi(kx) > k\phi(x).$$

This property is also known as strictly increasing returns to scale in the context of production functions. It is known that (for nonnegative functions on $[0, \infty)$ that vanish at 0) starshapedness lies between convexity and superadditivity (e.g., Bruckner and Ostrow, 1962). Because our definition of starshapedness involves a strict inequality, it lies between strict convexity and strict superadditivity.

For a_i , $i = 1, 2$, denote by a_{-i} the other action in A , that is, $a_{-1} = a_2$ and $a_{-2} = a_1$.

Definition 5. Under two actions, $A = \{a_1, a_2\}$, a solution $(m, (p_\beta)_\beta)$ to the principal's problem (P) is a **regret solution** if there exists a function $R: \mathbb{R} \rightarrow \mathbb{R}$ such that for all $(a_i, \omega) \in A \times \Omega$:

$$m(a_i, \omega) = R(v(a_i, \omega) - v(a_{-i}, \omega))$$

and

1. $R(x) \geq 0 \Leftrightarrow x \geq 0$,
2. $Q(x) := x + R(x) - R(-x)$ is starshaped.

Note that a regret solution does not only require that the agent's preferences can be represented by a utility function as in regret theory, but further that the emotional utility m itself can be written as the regret utility term in Bell (1982) and Loomes and Sugden (1982). That is because, like the emotional utility term in regret theory, m should be interpretable as experienced utility.

Property 1 states that the emotional utility is positive, 0, or negative if the counterfactual material payoff is worse, equal to, or better than the actual material payoff, respectively.¹² Property 2 requires that the sum of regret and rejoicing increases more than proportionally as a function of the material payoff difference. We also refer to it as disproportionate aversion to large regrets.

Proposition 1 and 2 below clarify the relation of our regret solution to original and generalized regret theory. While starshapedness is weaker than convexity, a regret solution still implies a generalized regret representation. Moreover, starshapedness is even sufficient to obtain the predictions of original regret theory that motivated the convexity assumption, namely the Allais paradox, the common-ratio effect, the isolation effect, and simultaneous gambling and insurance. As we show, starshapedness is in fact equivalent to the conjunction of these effects in the context of regret theory.

The following theorem shows that principal's problem indeed has a regret solution. This is perhaps surprising, since one could expect that the optimal emotional utility under some outcome depend on the whole decision problem. That is, the emotional utility could depend on the material payoff v of every outcome and on the prior. Instead, the emotional utility depends only on the difference between the realized material payoff and the material payoff had one taken the other action. Notably, the emotional utility does so following a regret-rejoicing function, which does not depend on the state nor on the prior. However, the regret-rejoicing function depends on λ and on the distribution of β , which we use for comparative statics below.

Theorem 1. Under a binary decision problem and Assumptions 1 to 3, the principal's problem (P) has a regret solution. The regret-rejoicing function R of such a solution is unique almost everywhere on $\mathbb{R}$.

¹²Loomes and Sugden (1982) make the stronger assumption that $R(x)$ is strictly increasing in x , that is they assume the better the actual compared to the counterfactual outcome, the better the emotion. The model of this paper does predict this property for negative x , but for large enough $x > 0$, $R(x)$ can be decreasing. In the context of the model, this can be interpreted as saying that for large utility differences, there is nothing to congratulate oneself for because any type would have paid enough attention to choose the right action with stakes that large.

The intuition for why regret and rejoicing are optimal is as follows. One can show that under exchangeability, the principal's problem separates by state, so the optimal emotions depend only on the material payoffs in the realized state. In any state, the agent's stochastic choice probabilities are given by a multinomial logit function (Matějka and McKay, 2015):

$$p_\beta(a_i|\omega) = \frac{e^{\frac{\beta}{\lambda}\Delta u(a_i,\omega)}}{1 + e^{\frac{\beta}{\lambda}\Delta u(a_i,\omega)}} \quad (4)$$

where u denotes the *emotion-inclusive utility*

$$u(a, \omega) := v(a, \omega) + m(a, \omega)$$

and

$$\Delta u(a_i, \omega) := u(a_i, \omega) - u(a_{-i}, \omega).$$

This illustrates that the optimal attention strategy is such that the probability with which the agent picks the right action in some state depends on the (emotion-inclusive) stakes in that state. The higher the stakes, the more attention the agent pays to make sure that he takes the right action under this state. What regret and rejoicing do is exactly to increase the stakes: regret penalizes the wrong action and rejoicing rewards the right action, state by state. This explains Property 1.

The solution satisfies starshapedness, or disproportionate aversion to large regrets, by the following. Recall that Q maps, in each state, the stakes without emotions Δv into the emotion-inclusive stakes $\Delta u = Q(\Delta v)$. If β was known,

$$Q(\Delta v) = \frac{\Delta v}{\beta},$$

would perfectly offset the agent's bias. Under unknown β , the optimal Q increases the stakes in some state to offset a *weighted average* of the bias. The average is taken with respect to the distribution of types weighted by how much these types' choice probabilities are *sensitive* to the stakes. Formally, the optimal $Q(\Delta v)$ is implicitly characterized by the ratio of v and an endogenous weighted average of the bias,

$$Q(\Delta v) = \frac{\Delta v}{\frac{\int p'_\beta(Q(\Delta v))^\beta dF(\beta)}{\int p'_\beta(Q(\Delta v)) dF(\beta)}}, \quad (5)$$

where

$$p'_\beta(Q(\Delta v)) = \frac{d}{dQ(\Delta v)} \frac{e^{\frac{\beta}{\lambda}Q(\Delta v)}}{1 + e^{\frac{\beta}{\lambda}Q(\Delta v)}},$$

For higher stakes, higher types are comparatively less sensitive to the stakes because they already take the right action with probability close to 1. Thus, the emotions are more targeted to the low types, which require proportionally higher emotions to offset their bias. This leads to disproportionate aversion to large regrets.

In general, equation (5) can have multiple solutions. We show that the solutions are increasing in the strong set order and thus wherever Q is non-unique it has an upward jump. The non-

decreasing function Q can have only countably many jump discontinuities, from which we show that R is unique up to at most countably many points.

Finally, the principal's problem can have solutions that are not regret solutions. When (5) has multiple solutions and there are duplicate states, she could pick from the solution set as a function of the state. Then, emotions are not a state-independent function of the set of payoff-differences anymore violating our regret solution definition. However, the regret solution is arguably the simplest type of solution because the principal can choose emotions only as a function of the vector of state-wise payoffs. In fact, she can employ the same function R *independent of the decision problem*.

Turning to the behavior induced by the optimal contract, it is useful to recall the distinction between the attention stage (choice of a signal structure) and the decision-making stage (choice of an action conditional on the signal). Corollary 1 shows that the principal only distorts behavior in the former stage.

Corollary 1. *The agent's behavior at the decision-making stage is not distorted. The agent's attention is skewed towards more extreme events.*

A common theme in the principal-expert literature is that the principal might want to distort the expert's incentives at the decision-making stage to incentivize more information acquisition, as already pointed out by Demski and Sappington (1987). In the current model, that is not necessary. Because of the symmetry implied by exchangeability, under any posterior that any type of the agent will optimally acquire, the action he takes is the same as the principal's preferred action, as we show in the Appendix. However, under other posteriors, regret and rejoicing can distort behavior, as is known from regret theory (Bell, 1982; Loomes and Sugden, 1982).

Regarding attention, starshapedness implies that the agent pays relatively more attention to more extreme states compared to the first-best. We measure extremeness of a state ω by the stakes $|v(a_1, \omega) - v(a_2, \omega)| = |\Delta v(a_1, \omega)|$. How much attention the type β agent pays to some state ω can be measured by the log-likelihood ratio of the signal under ω . Because the signal is identical to the action, equation (4) implies that the log-likelihood ratio equals

$$\log \left(\frac{p_\beta(a_1|\omega)}{p_\beta(a_2|\omega)} \right) = \log \left(e^{\frac{\beta}{\lambda} Q(\Delta v(a_1, \omega))} \right) = \frac{\beta}{\lambda} Q(\Delta v(a_1, \omega)).$$

By contrast, the log-likelihood ratio with respect to the first-best attention strategy p^* is

$$\log \left(\frac{p^*(a_1|\omega)}{p^*(a_2|\omega)} \right) = \log \left(e^{\frac{1}{\lambda} \Delta v(a_1, \omega)} \right) = \frac{1}{\lambda} \Delta v(a_1, \omega).$$

The ratio of those two log-likelihood ratios, whether for a fixed type β or in expectation over β , grows with the extremeness of the state, since $Q(\Delta v(a_1, \omega))/\Delta v(a_1, \omega)$ is increasing as $|\Delta v(a_1, \omega)|$ increases starshapedness. Thus, the model predicts that the agent pays relatively more attention to more extreme events.

The prediction of disproportionate attention to states with a high payoff difference is similar in spirit to models of salience or focusing based on contrast (Bordalo, Gennaioli, and Shleifer, 2012; Kőszegi and Szeidl, 2013). Salience effects or focusing are often understood as a feature of automatic bottom-up attention instead of deliberate top-down attention, such as rational inattention (Bordalo, Gennaioli, and Shleifer, 2022). In contrast, we show that such focusing effects are not only consistent with a model of rational (in)attention, but that they are in fact optimal distortions to motivate rational inattention.¹³

Our model makes two additional predictions regarding emotions that we relate to regret theory. First, while not imposed in standard regret theory, it is widely assumed that regret is stronger than rejoicing (e.g., Larrick and Boles, 1995; Humphrey, 2004), for which there is evidence based on self-reported emotions (Beattie, Baron, Hershey, and Spranca, 1994; Mellers, Schwartz, and Ritov, 1999). Moreover, models of regret often do not include rejoicing at all (Sarver, 2008), suggesting that regret is considered more prominent than rejoicing.

Corollary 2. *Regret is stronger than rejoicing, that is, $\forall x > 0: |R(-x)| > |R(x)|$.*

This result stems from the principal’s motive to minimize the cost of inducing emotions Ψ . Rejoicing is costly for the principal when the agent takes the right action, and regret is costly for the principal when the agent takes the wrong action. By the formula for conditional choice probabilities (4), the agent takes, in any state, the right action with higher probability than the wrong action. Thus, the principal finds it cheaper to use regret rather than rejoicing. This motive is present in the analysis of optimal moral systems by Kaplow and Shavell (2007), who assume an inculcation cost for guilt and virtue. Further, it echoes similar motives elsewhere in game theory as neatly expressed by Schelling (1980), “a promise is costly when it succeeds, and a threat is costly when it fails.”¹⁴

Second, in contrast to existing models of regret, our model endogenizes the regret as a function of the parameter λ , which can be interpreted as the difficulty of the decision problem. The higher λ the more costly it is to learn about the state and find the right action. Sugden (1986) argues that a short-coming of standard regret theory is that it does not incorporate that regret depends on how easy it was that one could have known the right action. He relates this to what he calls the self-recrimination aspect of regret. The following corollary shows that our model predicts stronger regret if it was cheaper to find out the optimal choice.

Corollary 3. *A smaller λ leads to larger regret from taking the wrong action, that is higher $|R(-x)|$ for all $x > 0$. Further, it leads to higher emotion-inclusive stakes, that is higher $Q(x)$ for all $x > 0$.*

Two mechanisms contribute to this effect. First, a smaller λ makes the types with lower β comparatively more responsive to changes in stakes, which increases $Q(x)$ and second, a smaller

¹³Lieder, Griffiths, and Hsu (2018) offer a different explanation for overrepresentation of extreme events based on the resource-rationality approach.

¹⁴Note that Corollary 2 does not necessarily imply that emotions are negative in expectation. In fact, for a quadratic cost of emotions $\psi(x) = x^2$, one can show that emotions are 0 in expectation.

λ makes it more likely that the agent takes the right action, which makes regret comparatively cheaper than rejoicing for the principal.

Corollary 3 can also be related to inter-personal differences. The parameter λ not only captures properties of the decision problem but also of the agent, namely how costly it is for them to process information. A small λ corresponds to decision-makers for whom it is relatively cheap to process information or, equivalently, who have a greater information processing capacity, for example due to a smaller cognitive load. The model predicts that they will experience stronger regret when taking the wrong action. This can help explain why Schwartz et al. (2002) and Iyengar, Wells, and Schwartz (2006) find that “maximizers” are less satisfied with their choices than “satisficers” despite the first group making better choices on average.

A.3 Relation to Classic Regret Theories

Next, we show that our model of regret lies between original and generalized regret theory. Nevertheless, it shares many of the predictions of original regret theory.

Regret theory was developed for decisions without taking into account attention. In particular, it has been applied to decisions between lotteries, where the nature of uncertainty is such that the state cannot be learned by paying attention. One way to reconcile this with our model is to assume that the agent’s internal self-control mechanism treats those decisions *as if* the state could be learned and triggers regret and rejoicing regardless.¹⁵

To compare our regret theory to original and generalized regret theory, we first define what it means for preferences to have regret representation as provided by our model, which we call an **evolutionary regret representation**.

Definition 6. *The preferences between actions have an evolutionary regret representation if there is a payoff function $v : A \times \Omega \rightarrow \mathbb{R}$ and function $Q : \mathbb{R} \rightarrow \mathbb{R}$ that is skew-symmetric, strictly increasing, and starshaped such that for any two actions, a_i and a_j , a_i is weakly preferred to a_j if and only if (2) holds.*

This definition is weaker than original regret representation by requiring that Q is only star-shaped instead of convex. It turns out that because starshaped implies superadditive, an evolutionary regret representation is still stronger than generalized regret theory, as the following proposition shows.

Proposition 1.

*If preferences have an **original** regret representation, they have an **evolutionary** regret representation.*

*If preferences have an **evolutionary** regret representation, they have a **generalized** regret representation.*

¹⁵Regret and rejoicing can be understood as rational *rules*, that perform well in most environments, in which arguably some learning about the state is possible. See for example Heiner (1983), Aumann (1997), and Halevy and Feltkamp (2005) for related discussions on rule rationality.

Does an evolutionary regret representation give up predictive power compared to original regret theory? Loomes and Sugden (1982) motivate the *convex* regret-rejoicing function Q through its implication of the Allais paradox, the common-ratio effect, the isolation effect, and under linear utility over money, the reflection effect, and simultaneous gambling and insurance. They show that the reflection effect holds under regret theory independent of assumptions on Q and that the isolation effect is equivalent to the common-ratio effect. Thus, we focus on the (a) Allais paradox, (b) the common-ratio effect, and (c) simultaneous gambling and insurance below.¹⁶ Proposition 2 shows that starshapedness is in fact equivalent to the conjunction of (a) to (c) in the context of regret theory.

The above-mentioned behavioral effects are formulated in terms of lotteries. To identify lotteries with actions in our present framework, let $x(a, \omega)$ denote the monetary outcome of action a under state ω . Given the prior μ over states, an action induces a lottery over monetary amounts $x(a, \omega)$. Following Loomes and Sugden (1982), lotteries can be identified with any action that induces such lottery given the belief μ . Two independent lotteries can be identified with two actions such that the two distributions over monetary amounts induced by the prior are independent. Let $(x_1, p; x_2, 1 - p)$ denote a lottery that gives monetary outcome x_1 with probability p and monetary outcome x_2 with probability $1 - p$. Further, let (x, p) be shorthand for $(x, p; 0, 1 - p)$.

The following definitions are taken from Loomes and Sugden (1982).

Definition 7 (Allais Paradox). *Let $X_1 = (x_1, p_1; x_2, \alpha)$ and $X_2 = (x_2, p_2 + \alpha)$ be independent lotteries where $1 \geq p_2 > p_1 > 0$ and $(1 - p_2) \geq \alpha \geq 0$. If there exists some probability $\bar{\alpha}$ such that $X_1 \sim X_2$ when $\alpha = \bar{\alpha}$, then (i) if $x_1 > x_2 > 0$, then $\alpha < \bar{\alpha} \Rightarrow X_1 \succ X_2$ and $\alpha > \bar{\alpha} \Rightarrow X_1 \prec X_2$ and (ii) if $0 > x_2 > x_1$, then $\alpha < \bar{\alpha} \Rightarrow X_1 \prec X_2$ and $\alpha > \bar{\alpha} \Rightarrow X_1 \succ X_2$.*

Definition 8 (Common Ratio Effect). *Let $X_1 = (x_1, \lambda p)$ and $X_2 = (x_2, p)$ be independent lotteries, where $1 \geq p > 0$ and $1 > \lambda > 0$. If there exists some probability $\bar{p}$ such that $X_1 \sim X_2$ when $p = \bar{p}$, then (i) if $x_1 > x_2 > 0$, then $p < \bar{p} \Rightarrow X_1 \succ X_2$ and $p > \bar{p} \Rightarrow X_1 \prec X_2$ and (ii) if $0 > x_2 > x_1$, then $p < \bar{p} \Rightarrow X_1 \prec X_2$ and $p > \bar{p} \Rightarrow X_1 \succ X_2$.*

Definition 9 (Simultaneous Gambling and Insurance). *Let v be linear in money. Let $X_1 = (0, 1)$ and $X_2(x, p; -px/(1 - p), 1 - p)$ be independent lotteries where $0 < p < 1$ and $x > 0$. (i) For $p > 0.5$, $X_1 \succ X_2$, (ii) for $p < 0.5$, $X_1 \prec X_2$, and for (iii) for $p = 0.5$, $X_1 \sim X_2$.*

Proposition 2. *Suppose the decision-maker has preferences between lotteries given by (2). The decision-maker's preferences exhibit the Allais paradox, common-ratio effect, and simultaneous gambling and insurance if and only if $Q(x)$ is starshaped.*

¹⁶Bell (1982) observes that Property 2 is sufficient for simultaneous gambling and insurance and for rejection of probabilistic insurance. Loomes and Sugden (1987) demonstrate that superadditivity of regret, which we show below follows from starshapedness, predicts preferences reversals.

B More than Two Actions

As stressed in the literature, it is not trivial how to generalize regret theory to more than two actions (Loomes and Sugden, 1982, Loomes and Sugden, 1987, Quiggin, 1994). We define a *general regret solution* based on the utility representation in Loomes and Sugden (1987) and Sugden (1993), which to our knowledge accommodates all proposals. They define a representation of preferences where the utility depends on the realized outcome and on the set of counterfactual outcomes under the same state. They do not impose additional assumptions that would correspond to generalizations of Properties 1 and 2. In fact, it is not trivial how these would generalize to more than two actions. This representation is more general than another widely-used representation due to Quiggin (1994), according to which regret depends on the difference of the actual to the highest attainable payoff under the realized state. As we will show, that form of regret is in general not optimal in our model.

Definition 10 (General Regret Solution). *Under a finite choice set A , a solution $(m, (p_\beta)_\beta)$ to the principal's problem (P) is a general regret solution if there exists a function R such that for all $(a, \omega) \in A \times \Omega$:*

$$m(a, \omega) = R(\{v(a, \omega) - v(a', \omega)\}_{a' \in A \setminus \{a\}}),$$

and for all $a, a' \in A, \omega \in \Omega$:

$$v(a, \omega) > v(a', \omega) \Leftrightarrow v(a, \omega) + m(a, \omega) > v(a', \omega) + m(a', \omega).$$

The definition states that emotions are a state-independent function of the *set* of payoff differences between the actual and the counterfactual payoffs under the realized state. In particular, a permutation of the counterfactual payoffs does not change the emotion as well as shifting all payoffs under some state by a constant. Moreover, adding emotions preserves the strict ordering of payoffs, state-wise. In the Appendix B, we comment on how our representation relates exactly to the one in Loomes and Sugden (1987).

Finally, assume that the principal has strict preferences over actions, given the state. This can be interpreted as a genericity assumption or as the state being defined in a sufficiently fine-grained way. We use this assumption to prove that state-wise permutations of payoffs translate to corresponding permutations of the optimal emotions.

Assumption 4. *For all $\omega \in \Omega$, for all $a, a' \in A$: $(a \neq a' \rightarrow v(a, \omega) \neq v(a', \omega))$.*

Under these assumptions, we prove the following theorem.

Theorem 2. *Under finite A and Assumptions 1 to 3, the principal's problem (P) has a general regret solution.*

The main part of the proof is to show that the principal's problem separates by state and that state-wise optimal emotions exist, a result which we also use for proving Theorem 1. Moreover,

we show that state-wise optimal emotions are invariant under state-wise permutations or shifts of payoffs, and preserve strict orders of payoffs.¹⁷

Our model does not only predict that the optimal emotions are a general regret solution. Given λ and the distribution of β , the model can be used to compute the precise optimal emotional payoffs. While optimal emotions could, again, depend on the whole decision problem, they boil down to a state-wise $|A|$ -variable optimization problem. That is, the optimal emotions are a function of the vector of payoffs in the realized state. They are given by the cost-of-emotions-minimizing solution to

$$\max_{u(\cdot, \omega): A \rightarrow \mathbb{R}} \int \left(\lambda H(p_\beta(u(\omega))) + \sum_{a \in A} p_\beta(a|u(\omega)) v(a, \omega) \right) dF(\beta),$$

where $p_\beta(a|u(\omega))$ is given by the multinomial logit formula

$$p_\beta(a|u(\omega)) = \frac{e^{\frac{\beta}{\lambda} u(a, \omega)}}{\sum_{a' \in A} e^{\frac{\beta}{\lambda} u(a', \omega)}},$$

and $H(p_\beta(u(\omega)))$ is the entropy of the probability vector $p_\beta(u(\omega)) = (p_\beta(a|u(\omega)))_{a \in A}$. Unfortunately, the problem does not allow for a simple analytical solution or for comparative statics as in binary decision problems. However, once the problem is solved for a vector of payoffs, it applies to payoff-equivalent states in *any decision problem*.

IV Conclusion

While counterfactual thinking is often dismissed as irrational (e.g. Howard, 1992)¹⁸ and seen as distortions to rational decision-making, this paper shows that regret, which is based on counterfactual reasoning, is actually optimal to motivate better decision-making. In a binary decision problem, the model predicts disproportionate aversion to large regrets and that attention is skewed towards extreme states. Extending standard regret theory, the model predicts stronger regret the easier one could have known the right action, consistent with the psychology of a self-blaming aspect of regret.

Many open questions remain. Our analysis is for tractability and conciseness restricted by symmetry assumptions. Can the principal profit from asymmetric incentive schemes in symmetric decision problems? What are the optimal incentives when the underlying decision problem is asymmetric? When delegating asymmetric decision problems, motivating attention may come only at the cost of distorting decision-making. Analyzing the optimal incentives could help shed further light on the resulting trade-off and thereby unite themes from the current paper, which shows that regret is optimal to motivate attention absent distortions of choice, and studies in regret theory, which focus on how regret distorts decision-making.

¹⁷Solutions that are not general regret solutions exist for the same reason as in the case with two actions. Further, one can show analogously to the binary action case that the principal distorts behavior only at the attention stage and not at the decision-making stage.

¹⁸However, see Bradley and Stefánsson (2017) for a defense of counterfactual comparisons as rational.

On a broader scale, our research underscores parallels between internal psychological mechanisms, such as counterfactual comparisons, and more traditional principal-agent problems. We believe that such a perspective has the potential to profit both behavioral economics and mechanism design. Behavioral economics might glean insights from mechanism design to better understand the function and benefits of non-standard preferences. Similarly, mechanism design can turn to human psychology as a source of innovative solutions for agency problems, which evolution has already solved.

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V Appendix A: Proofs

A Lemma 1

A general mechanism elicits a message from the agent *before* as well as *after* attention. Conditional on the messages, an action recommendation is sent and emotion-transfers are assigned conditional on the messages and the outcome. By the revelation principle it is sufficient to restrict attention to mechanisms, where the first message corresponds to the agent’s type. We maintain the assumption that any emotion scheme induces an exchangeable decision problem for the agent. First, we analyze the case when the mechanism elicits a message only before attention. At the end of the proof, we argue that eliciting a message after attention can be reduced to a mechanism of the former case.

The principal chooses a vector of emotion schemes $m_\beta: A \times \Omega \rightarrow \mathbb{R}$ and attention strategies $p_\beta: \Omega \rightarrow \Delta(A)$, indexed by the agent’s type, each, subject to incentive compatibility, budget balance, and exchangeability. The incentive-compatibility constraint below captures both the truthful reporting of one’s type as well as the subsequent attention and decision-making, accounting for the possibility of double deviations. The principal’s problem is

$$\begin{aligned}
& \max_{(m_\beta, p_\beta)_\beta} \int \left(-c(p_\beta) + \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p_\beta(a|\omega) v(a, \omega) \right) dF(\beta) \\
& \quad \text{s.t.} \\
& \forall \beta: (\beta, p_\beta) \in \arg \max_{\hat{\beta}, \hat{p}} -c(\hat{p}) + \beta \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) \hat{p}(a|\omega) (v(a, \omega) + m_{\hat{\beta}}(a, \omega)) \quad (\text{IC}) \\
& \int \left(\sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p_\beta(a|\omega) m_\beta(a, \omega) \right) dF(\beta) = 0 \quad (\text{BC}) \\
& \forall \beta: v + m_\beta \text{ is exchangeable.} \quad (\text{EX})
\end{aligned}$$

Out of the solutions $(m_\beta, p_\beta)_\beta$, the principal chooses the one that minimizes the variance of emotions,

$$\int \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p_\beta(a|\omega) m_\beta(a, \omega)^2 dF(\beta).$$

Osband (1989) shows that in their model offering a menu of emotion schemes can in principle benefit the principal, but using numerical simulations he shows that these gains are very small.

In contrast to Osband, in our setup, heterogeneity concerns the agent's bias instead of his cost parameter. Thus, the first-best attention strategy is the same for every agent, which the proof relies on. In Yoder (2022), the principal can effectively screen agents when the acquired posterior is contractible. We show that the hidden-action constraint implies that the principal can induce only gross value functions that are convex in the posterior. Offering a menu makes the gross value function more convex, leading to further distortions at the top and bottom. Our proof does not rely on the specific shape of mutual information and we show the proposition for any posterior-separable attention cost (see section VI for a definition) with strictly concave and differentiable H that is *symmetric in the states*.

Proof. Under exchangeability of v , if there are two states, $\Omega = \{\omega_1, \omega_2\}$, then one of the following needs to be the case. The payoff v is state-wise constant, or $\mu(\omega_1) = \mu(\omega_2) = \frac{1}{2}$ and there are two actions, $A = \{a_1, a_2\}$, with exchangeable payoffs, $v(a_1, \omega_1) = v(a_2, \omega_2)$ and $v(a_1, \omega_2) = v(a_2, \omega_1)$. If there were more than two actions and v is not state-wise constant, then more than two states are necessary to have all permutations of $(v(a, \omega))_{a \in A}$ be equiprobable.

If the payoffs are state-wise constant, then the optimal attention strategy is to pay zero attention, which is achieved at the smallest implementation cost by setting the emotions to 0 for all types. Thus, all u_β are the same.

Consider, then, the other case that or $\mu(\omega_1) = \mu(\omega_2) = \frac{1}{2}$ and there are two actions, $A = \{a_1, a_2\}$, with exchangeable payoffs, $v(a_1, \omega_1) = v(a_2, \omega_2)$ and $v(a_1, \omega_2) = v(a_2, \omega_1)$. Without loss, the payoff matrix for the principal can be written as

	ω_1	ω_2
a_1	v	0
a_2	0	v

where $v > 0$.

Agent's Behavior Turning to the agent, we first analyze the agent's behavior given some $u: A \times \Omega \rightarrow \mathbb{R}$ in the menu of utility functions $(u_\beta)_\beta = (v + m_\beta)_\beta$ that the principal offers.¹⁹ By the exchangeability and an argument analogous to the above, the payoff matrix of any such u ,

	ω_1	ω_2
a_1	$u(a_1, \omega_1)$	$u(a_1, \omega_2)$
a_2	$u(a_2, \omega_1)$	$u(a_2, \omega_2)$

must satisfy one of the following.

Either u is state-wise constant, that is $u(a_1, \omega_1) = u(a_2, \omega_1)$ and $u(a_2, \omega_1) = u(a_1, \omega_1)$, or $u(a_1, \omega_1) = u(a_2, \omega_2)$ and $u(a_1, \omega_2) = u(a_2, \omega_1)$. In the first case, we can make u constant over all outcomes without changing the agent's behavior (he pays no attention in either case), so it is

¹⁹By a change of variables, the principal chooses u directly instead of m .

captured by the second case, which we analyze below. We can write the agent's payoff matrix as

	ω_1	ω_2
a_1	$\underline{u} + \Delta u$	$\underline{u}$
a_2	$\underline{u}$	$\underline{u} + \Delta u$

where $\underline{u} = u(a_1, \omega_2) = u(a_2, \omega_1)$ and $\Delta u = u(a_1, \omega_1) - u(a_2, \omega_1) = u(a_2, \omega_2) - u(a_1, \omega_2)$.

We characterize the optimal attention strategy given some chosen utility function u . Following Caplin and Dean (2013), we solve for the optimal distribution over posteriors, induced by the attention strategy, analogous to Bayesian persuasion models (Kamenica and Gentzkow, 2011). Under two states, we can identify posteriors $\gamma \in \Delta(\Omega)$ with the probability in $[0, 1]$ they assign to state ω_1 , which makes this posterior approach especially attractive. The problem of agent β given u is to pick a Bayes-consistent distribution over posteriors to maximize the expectation of the *net value* function N_β (Caplin and Dean, 2013):

$$\begin{aligned} \max_{\tau \in \Delta[0,1]} \quad & \mathbb{E}_\tau[N_\beta(\gamma|u)] \\ \text{s.t.} \quad & \\ & \mathbb{E}_\tau[\gamma] = \frac{1}{2} \end{aligned}$$

Here, $N_\beta(\gamma|u)$ is the net value of posterior γ for the agent of type β given u , which is

$$N_\beta(\gamma|u) = \beta(\underline{u} + \max\{\gamma\Delta u, (1 - \gamma)\Delta u\}) + \lambda H(\gamma).$$

The first summand of the net value $N_\beta(\cdot|u)$ captures the *gross value*, or instrumental value, of attention and the second the attention cost. The gross value is continuous, convex, symmetric around $1/2$, and piecewise linear as the maximum of linear functions. The attention cost is mutual information, $H(\gamma) = \gamma \log(\gamma) + (1 - \gamma) \log(1 - \gamma)$, which is strictly concave, differentiable and symmetric around $1/2$. These are the only properties of H that we make use of for the proof.

As known from the literature on Bayesian persuasion, the solution can be obtained by concavifying the net value function $N_\beta(\cdot|u)$. As we show next, this Bayesian persuasion problem takes a particularly simple form due to symmetry.

The net value function has one global maximum if $\Delta u = 0$ and two global maxima otherwise. Restricted to $[1/2, 1]$, the net value function attains its maximum by continuity. On $[1/2, 1]$, the net value function is the sum of the linear gross value and the strictly concave attention cost, so the maximum must be unique. By symmetry and differentiability of H , we have $H'(1/2) = 0$, so the maximum is at $1/2$ only if $\Delta u = 0$. In fact, by differentiability the maximum is characterized by $-\lambda H'(\gamma) = \beta \Delta u$ if $-\lambda H'(1) > \beta \Delta u$ and $\gamma = 1$ otherwise. By symmetry of H around $1/2$, the net value function $N_\beta(\cdot|u)$ is symmetric around $1/2$, so the net value function attains another maximum at the reflection point around $1/2$.

Because the maxima are symmetric around the prior probability $1/2$, there is a unique Bayes-

consistent τ whose support is the set of global maxima. This is the unique τ that attains the highest expected posterior net value because each posterior in its support maximizes the net value function. This τ is either degenerate at $1/2$ or binary and symmetric around $1/2$. The agent selects action a_1 when the realized posterior is greater than $1/2$ and a_2 when the realized posterior is less than $1/2$, if $\Delta u > 0$ and vice versa if $\Delta u < 0$. Thus, given $v > 0$, the principal never offers an emotion-inclusive utility with $\Delta u < 0$, which would induce the agent to pay costly attention only to select the wrong action.

Thus, while the agent chooses a high-dimensional distribution over posteriors, we can characterize his choice by the one-dimensional maximizer of the net value function on $[1/2, 1]$, which corresponds to the higher posterior the agent acquires. As written above, the maximizer is characterized by $\lambda - H'(\gamma) = \beta \Delta u$ if $-\lambda H'(1) > \beta \Delta u$ and $\gamma = 1$ otherwise. Because H is strictly concave and symmetric, $-H'$ is strictly increasing on $[1/2, 1]$ with $-H'(1/2) = 0$. Thus, the maximum is strictly increasing in Δu as long as the optimal γ is less than 1 and constant afterwards.

The complete strategy of the agent consists of choosing an emotion scheme and subsequently an attention strategy. By the above, the agent's behavior can be described as choosing a net value function and subsequently the maximum of the net value function on $[1/2, 1]$. This is equivalent to choosing the maximum on $[1/2, 1]$ of the pointwise maximum $N_\beta(\cdot | (u_\beta)_\beta)$ of the collection of net value functions,

$$\begin{aligned} N_\beta(\gamma | (u_\beta)_\beta) &= \max_{\hat{\beta}} \{N_\beta(\gamma, u_{\hat{\beta}})\} \\ &= \beta \max_{\hat{\beta}} \{u_{\hat{\beta}} + \gamma \Delta u_{\hat{\beta}}\} + \lambda H(\gamma), \text{ for } \gamma \geq 1/2. \end{aligned}$$

We call $N_\beta(\gamma | (u_\beta)_\beta)$ the *effective* value function. By $\Delta u_\beta > 0$ for all β , the effective value function is supermodular on $[1/2, 1]$ in β and γ and thus the optimal γ is non-decreasing in β (Topkis, 1978). The effective value function on $[1/2, 1]$ is the sum of the maximum of increasing, linear functions and a concave attention cost. The maximum of increasing, linear functions is increasing and convex. Thus, non-decreasing γ implies a non-decreasing choice of the slope $\Delta u_{\hat{\beta}}$.

To summarize, for any type of the agent, the menu of emotion-inclusive utility functions induces an effective value function and the type's acquired signal structure can be described as picking the posterior in $[1/2, 1]$ that maximizes this net value function. We have shown that (1) a higher β implies a weakly more extreme posterior, and (2) a weakly higher slope Δu_β of the gross value function. Given the choice of an emotion-inclusive utility, (3) the agent's posterior is a strictly increasing function of the slope of the gross value function, if the posterior is below 1.

Principal's Problem Turning back to the principal, her net value function is

$$N_P(\gamma) = \max\{\gamma v, (1 - \gamma)v\} + \lambda H(\gamma).$$

Her value from the symmetric distribution with support $\{\gamma, 1-\gamma\}$ is just $N_P(\gamma)$ by symmetry of N_P . The principal's value function N_P is strictly concave on $[1/2, 1]$ because it is the sum of the linear gross value and the strictly concave attention cost. Strict concavity implies strict quasi-concavity, which we are using below.

Suppose for the sake of contradiction that there are types β, β' such that $u_\beta \neq u_{\beta'}$. We show that the principal strictly benefits from offering only the emotion-inclusive utility u_{β^*} of a type β^* that selects the most favorable posterior to the principal,

$$\beta^* \in \arg \max_{\beta \in \text{supp}(\beta)} N_P(\gamma_\beta).$$

Such a type β^* exists by $\text{supp}(\beta)$ being finite.

By (2), greater types than β^* select a weakly higher slope and smaller types a weakly smaller slope of the gross value function. Offering only the emotion-inclusive utility u_{β^*} , by (3) weakly increases the posterior for type less than β^* , weakly decreases the posterior for types greater than β^* , and does not change the posterior of β^* . Because the ordering of posteriors, (1), still holds due to supermodularity of the new value function, each posterior moves weakly towards γ_{β^*} . By strict quasi-concavity of the principal's value function on $[1/2, 1]$ and γ_{β^*} yielding a weakly higher value to the principal, the principal benefits weakly from the induced changes and benefits strictly from strict changes. Thus, the principal is weakly better off offering only one emotion scheme.

It remains to show that in case of indifferences, the principal cannot decrease the implementation cost by offering different emotion schemes. By the above, the principal is strictly better off if some posterior moved strictly. The only case that no posterior changed, is when the only types whose emotion-inclusive utility changed selects $\gamma = 1$ at their original and at the new emotion-inclusive utility. By construction, the new value function is below the original induced value function at $\gamma = 1$, so it entails a smaller emotional payoff after selecting the better action under the realized state. Because $\beta \leq 1$, optimality entails that the slope of the constructed gross value function is at least v . Then, the implementation cost being minimized entails positive emotions at selecting the better action under the realized state. Thus, the new emotion-inclusive utility assigning smaller emotions implies a reduction of the implementation cost.

General Mechanism As mentioned above, the principal could use an even more general mechanism. That is, she could elicit an additional message from the agent after attention and condition the emotion scheme on both the message before and after attention. We maintain the restriction that any assigned emotion scheme m is such that $v + m$ is exchangeable. Here, we sketch how the proof can be extended to account for such mechanisms. We show that by starting with a general mechanism, we can construct a mechanism *without* a message after attention that is equally good for the principal. Then, the above proof can be applied. The argument below uses that if the agent acquires symmetric posteriors, there is no difference between eliciting a message before and after attention, as we have shown above for symmetric value functions. Using this, we show that the

principal might as well offer only one emotion scheme given the first message from the agent (thus making the second message meaningless). The argument proceeds in three steps.

First, given an incentive-compatible general mechanism, fix an arbitrary type β of the agent and consider the agent's attention problem following the first message. We can, again, by direct revelation identify the first message with the agent's type, which the agent truthfully reveals. The agent's attention problem still consists of concavifying a symmetric value function. That is because analogous to the argument above, the agent faces an effective value function, denoted by $\hat{N}_\beta$, given by the maximum of the value functions induced by the emotion schemes, which the agent can choose from after attention. Because each emotion scheme induces a symmetric value function, the maximum of the value functions is symmetric, too. This implies that the set of maximizers M_β of the value function $\hat{N}_\beta$ is symmetric around $1/2$. Any Bayes-consistent distribution with support at only the maximizers of the $\hat{N}_\beta$ is a solution to the attention problem of type β .

Second, there is a symmetric, principal's preferred solution to the attention problem of type β . The principal's value function is symmetric around $1/2$, too. Hence, the principal prefers, out of the solutions to the agent's problem, those whose support includes only posteriors that give the highest value to the principal. Formally, let $M_{P,\beta} := \arg \max_{\gamma \in M_\beta} N_P(\gamma)$, which is symmetric around $1/2$. The principal's preferred solutions to the attention problem of type β are the Bayes-consistent distributions with support in $M_{P,\beta}$. Because $M_{P,\beta}$ is symmetric and the prior is at $1/2$, there is a Bayes-consistent, binary, symmetric distribution with support in $M_{P,\beta}$.

Third, starting with a mechanism that elicits a message before and after attention, we can construct an equally good mechanism that elicits a message only before attention. As we are allowing the principal to pick her preferred solution when the agent is indifferent, for any type we can select a principal's preferred solution to his attention problem that is moreover symmetric and binary by the above. By symmetry, this solution makes any agent pick the same emotion scheme (or, send the same second message) after each acquired posterior. This solution can also be implemented by offering to every agent *only this one emotion scheme*, after the first message. Offering less choice to the agent does not violate any other incentive-compatibility constraints. Finally, we can apply the proof above to the resulting mechanism. $\square$

B Theorem 2

Before proving Theorem 2, which we use below to prove Theorem 1, we comment on how our general regret representation relates to the one in Loomes and Sugden (1987).

To introduce the general regret representation in Loomes and Sugden (1987), let us define $o(a, \omega)$ as the outcome from action a under state ω . Their representation requires that the agent maximizes a utility function

$$u(a, \omega) = \Phi(o(a, \omega), \{o(a', \omega) | a \in A \setminus \{a\}\}),$$

that is a function of the actual outcome and the set of counterfactual outcomes.

Our general regret solution

$$u(a, \omega) = v(a, \omega) + R(\{V(a, \omega) - v(a', \omega) | a' \in A \setminus \{a\}\})$$

implies their representation but makes several additional assumptions. First, the emotion-inclusive utility $u(a, \omega)$ depends on the set of state-wise *payoffs* instead of outcomes. Thus, payoff-equivalent states are treated the same way. Second, the additive separability together with the dependence of emotions on the set of payoff *differences* additionally imposes that shifting all payoffs in some state by a constant does not change the emotions. Third, we assume that emotions do not change, in each state, the strict ordering of utilities, as in Sarver (2008). These three additional assumptions seem plausible in the context of regret theory. We now turn to the proof of Theorem 2.

Proof: Let $A := \{a_1, \dots, a_n\}$. Because the principal is constrained to choose emotions that induce an exchangeable emotion-inclusive utility u , the state-dependent choice probabilities of any type of the agent can be written as (Matějka and McKay, 2015)

$$p_\beta(a|\omega) = \frac{e^{\frac{\beta}{\lambda}u(a,\omega)}}{\sum_{a' \in A} e^{\frac{\beta}{\lambda}u(a',\omega)}} = \frac{e^{\frac{\beta}{\lambda}\Delta u(a,\omega)}}{1 + \sum_{a' \neq a_1} e^{\frac{\beta}{\lambda}\Delta u(a',\omega)}}. \quad (6)$$

where $\Delta u(a, \omega) := u(a, \omega) - u(a_1, \omega)$. Further, we define $u(\omega) := (u(a_1, \omega), \dots, u(a_n, \omega))$ and $p_\beta(\omega) := (p_\beta(a_1|\omega), \dots, p_\beta(a_n|\omega))$.

By (6), the choice probabilities only depend on the state-wise utilities and even further only on state-wise utility differences with respect to some reference action. Using this, we show that the principal's problem is separable by state.

By a change of variables, the principal chooses directly the emotion-inclusive utility u instead of emotions m . The principal's maximization problem consists of an outer minimization problem regarding the implementation cost $\psi(u(a, \omega) - v(a, \omega))$ and an inner problem to maximize the payoff net of attention cost, which is subject to the exchangeability constraint:

$$\min_{u: A \times \Omega \rightarrow \mathbb{R}} \int \left(\sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p_\beta(a|\omega) \psi(u(a, \omega) - v(a, \omega)) \right) dF(\beta)$$

s.t.

$$u \in \arg \max_{u: A \times \Omega \rightarrow \mathbb{R}} \int \sum_{\omega \in \Omega} \mu(\omega) \left(\lambda \log(1/n) + \lambda H(p_\beta(\omega)) + \sum_{a \in A} p_\beta(a|\omega) v(a, \omega) \right) dF(\beta) \quad (7)$$

s.t. $\{a_1, \dots, a_n\}$ is exchangeable with respect to u .

where $p_\beta(a|\omega)$ is given by (6) and $H(p_\beta(\omega)) = \sum_{a \in A} -p_\beta(a|\omega) \log(p_\beta(a|\omega))$ is the entropy of the conditional choice probabilities. We have used that under exchangeable payoffs, the unconditional choice probabilities are all the same, and thus equal $1/n$. The constant entropy of the unconditional choice probabilities, $\lambda \log(1/n)$, can be dropped.

Inner Maximization Problem Turning to the inner maximization problem first, we can exchange the order of integration and summation in (7). We first relax the maximization problem by dropping the exchangeability constraint. Then, the inner objective is separable by state as $p_\beta(\omega)$ depends only on the vector of state-wise utilities, $u(\omega)$. Below, we show that any solution to the state-wise problem

$$\max_{u(\cdot, \omega): A \rightarrow \mathbb{R}} \int \left(\lambda H(p_\beta(\omega)) + \sum_{a \in A} p_\beta(a|\omega) v(a, \omega) \right) dF(\beta) \quad (8)$$

preserves the strict order of utilities and that a solution exists. Then, we go on to show that there is an exchangeable utility function that maximizes (8) state-by-state. By every state having positive probability, this implies that any solution to the constrained inner-maximization problem (including the exchangeability constraint) must maximize the state-wise problem in every state.

Lemma 2. *Any solution to the state-wise maximization problem (8) satisfies*

$$\forall a, a' \in A: v(a, \omega) > v(a', \omega) \Rightarrow u(a, \omega) > u(a', \omega).$$

Proof. If $v(a, \omega) > v(a', \omega)$ but $u(a, \omega) < u(a', \omega)$, then we can improve the principal's utility by exchanging $u(a, \omega)$ and $u(a', \omega)$, that is setting $u(a, \omega)$ to the original value of $u(a', \omega)$ and vice versa. For any type β , this would not change the entropy term while improving the expected payoff v . That is because for any type β , exchanging these two u 's only exchanges the probabilities $p_\beta(a|\omega)$ and $p_\beta(a'|\omega)$ due to (6). The entropy $H(p_\beta(u(\omega)))$ remains unchanged for any type β because it is invariant under permutations of the probabilities, which is immediate from the definition of entropy. The expected payoff $\sum_{\hat{a} \in A} p_\beta(\hat{a}|\omega) v(\hat{a}, \omega)$ would be improved because the conditional choice probabilities of $v(a, \omega)$ and $v(a', \omega)$ are exchanged, giving higher probability to the higher payoff. Hence, at a maximum, $v(a, \omega) > v(a', \omega)$ implies $u(a, \omega) \geq u(a', \omega)$.

Suppose for the sake of contradiction that $v(a, \omega) > v(a', \omega)$ and $u(a, \omega) = u(a', \omega)$. Set $\tilde{u}(a, \omega) = u(a, \omega) + \varepsilon$ and $\tilde{u}(a', \omega) = u(a', \omega) - \varepsilon$ and $\tilde{u}(\hat{a}, \omega) = u(\hat{a}, \omega)$ for all $\hat{a} \neq a, a'$. Fix any type $\beta \in (0, 1]$. At $\varepsilon = 0$, $p_\beta(a|u(\omega)) = p_\beta(a'|u(\omega))$ by (6). By symmetry of (6) in the u 's, at $\varepsilon = 0$, for all $\hat{a} \neq a, a'$: $\frac{dp_\beta(\hat{a}|\omega)}{du(a, \omega)} = \frac{dp_\beta(\hat{a}|\omega)}{du(a', \omega)}$ and hence $\frac{dp_\beta(\hat{a}|\omega)}{d\varepsilon} = \frac{dp_\beta(\hat{a}|\omega)}{du(a, \omega)} - \frac{dp_\beta(\hat{a}|\omega)}{du(a', \omega)} = 0$ and $\frac{dp_\beta(a|\omega)}{d\varepsilon} = -\frac{dp_\beta(a'|\omega)}{d\varepsilon}$. By symmetry of entropy in the conditional choice probabilities, at $\varepsilon = 0$, $\frac{dH}{dp_\beta(a|\omega)} = \frac{dH}{dp_\beta(a'|\omega)}$ and hence $\frac{dH}{d\varepsilon} = \frac{dp_\beta(a'|\omega)}{d\varepsilon} \frac{dH}{dp_\beta(a'|\omega)} + \frac{dp_\beta(a|\omega)}{d\varepsilon} \frac{dH}{dp_\beta(a|\omega)} = 0$ by the equalities above. However, $v(a, \omega) - v(a', \omega) > 0$, so at $\varepsilon = 0$, the derivative of the expected payoff is positive by $\forall \hat{a} \neq a, a': \frac{dp_\beta(\hat{a}|\omega)}{d\varepsilon} = 0, \frac{dp_\beta(a|\omega)}{d\varepsilon} > 0$ and $\frac{dp_\beta(a'|\omega)}{d\varepsilon} < 0$. Hence, increasing ε by some positive amount improves the principal's utility for all types β , which is why we must have $u(a, \omega) > u(a', \omega)$. $\square$

Because of the genericity Assumption 4, the Lemma implies that

$$v(a, \omega) > v(a', \omega) \Leftrightarrow u(a, \omega) > u(a', \omega).$$

Before we show existence, we insert the agent's state-dependent choice probabilities (6) to

simplify the integrand of the state-wise objective:

$$\begin{aligned}
& \sum_{a \in A} \left(p_\beta(a|\omega) v(a, \omega) \right) + \lambda H(p_\beta(u(\omega))) \\
&= \sum_{a \in A} \left(p_\beta(a|\omega) v(a, \omega) - \lambda p_\beta(a|\omega) \log(p_\beta(a|\omega)) \right) \\
&= \sum_{a \in A} \left(p_\beta(a|\omega) v(a, \omega) - \lambda p_\beta(a|\omega) \log \left(\frac{e^{\frac{\beta}{\lambda} u(a, \omega)}}{\sum_{a' \in A} e^{\frac{\beta}{\lambda} u(a', \omega)}} \right) \right) \\
&= \sum_{a \in A} \left(p_\beta(a|\omega) (v(a, \omega) - \beta u(a, \omega)) + \lambda \log \left(\sum_{a' \in A} e^{\frac{\beta}{\lambda} u(a', \omega)} \right) \right)
\end{aligned}$$

Thus, the state-wise objective, which we denote by $G(u(\omega))$, is

$$G(u(\omega)) = \int \left(\sum_{a \in A} p_\beta(a|u(\omega)) (v(a, \omega) - \beta u(a, \omega)) + \lambda \log \left(\sum_{a \in A} e^{\frac{\beta}{\lambda} u(a, \omega)} \right) \right) dF(\beta). \quad (9)$$

It will be useful to know the derivatives of G and $p_\beta(a|\omega)$ with respect to the components of u , which have simple expressions. Using

$$p_\beta(a|\omega) = \frac{d}{du(a, \omega)} \frac{\lambda}{\beta} \log \left(\sum_{a' \in A} e^{\frac{\beta}{\lambda} u(a', \omega)} \right),$$

which follows from (6), one can show by straightforward differentiation

$$\frac{dG(u(\omega))}{du(a, \omega)} = \int \left(\sum_{a' \in A} \frac{dp_\beta(a'|\omega)}{du(a, \omega)} (v(a', \omega) - \beta u(a', \omega)) \right) dF(\beta) \quad (10)$$

$$\frac{dp_\beta(a|\omega)}{du(a, \omega)} = \frac{\beta}{\lambda} p_\beta(a|\omega) (1 - p_\beta(a|\omega)) = \frac{\beta}{\lambda} p_\beta(a|\omega) \sum_{a' \neq a} p_\beta(a'|\omega) \quad (11)$$

$$\frac{dp_\beta(a'|\omega)}{du(a, \omega)} = -\frac{\beta}{\lambda} p_\beta(a'|\omega) p_\beta(a|\omega), \text{ for } a \neq a'. \quad (12)$$

Lemma 3. *The state-wise maximization problem (8) has a solution.*

Proof. We prove the statement by showing that any $u(\cdot, \omega) \in \mathbb{R}^A$ is weakly dominated (in the sense of achieving a lesser or equal objective) by some $\tilde{u}(\cdot, \omega) \in \mathbb{R}^A$ in a fixed compact subset of $\mathbb{R}^A$. Thus, we can restrict attention to this compact set, on which a maximum (which must be a global maximum) is obtained by continuity of the objective.

Without loss, assume $v(a_1, \omega) \leq \dots \leq v(a_n, \omega)$. We can also assume without loss that $v(a_1, \omega) = u(a_1, \omega) = 0$. That is because from the state-wise objective (8) it is clear that shifting all $u(a, \omega)$ by the same constant does not affect choice probabilities and hence also not the objective. Similarly, shifting all $v(a, \omega)$ by some constant c changes the objective exactly by that constant since $\sum_{a \in A} p_\beta(a|\omega) (v(a, \omega) + c) = c + \sum_{a \in A} p_\beta(a|\omega) v(a, \omega)$. Hence, the constant does not affect the set

of maximizers.

By Lemma 3, if for some a, a' , $v(a, \omega) > v(a', \omega)$ and $u(a', \omega) > u(a, \omega)$, then we can exchange $u(a, \omega)$ and $u(a', \omega)$ and improve the principal's utility. If $v(a, \omega) = v(a', \omega)$ and $u(a', \omega) > u(a, \omega)$, then exchanging $u(a, \omega)$ and $u(a', \omega)$ does not change the principal's utility. Thus, every u that does not satisfy $u(a_1, \omega) \leq \dots \leq u(a_n, \omega)$ is weakly dominated by a $\tilde{u}$ that does satisfy it. Thus, we can restrict attention to the set of utility-vectors u with $0 = u(a_1, \omega) \leq \dots \leq u(a_n, \omega)$.

Finally, we bound $u(a_i, \omega)$ for $i = 2, \dots, n$ from above by $u(a_{i-1}, \omega) + \frac{v(a_n, \omega)}{\inf \beta}$ ($\frac{1}{\inf \beta}$ exists by Assumption 1). Suppose for some $m \in \{2, \dots, n\}$, $u(a_m, \omega)$ violated the bound. Then, for all $i \geq m, k \leq m-1$:

$$u(a_i, \omega) \geq u(a_m, \omega) > u(a_{m-1}, \omega) + \frac{v(a_n, \omega)}{\inf \beta} \geq u(a_k, \omega) + \frac{v(a_n, \omega)}{\inf \beta}. \quad (13)$$

Using (10) through (12), the derivative of decreasing $u(a_m, \omega)$ through $u(a_n, \omega)$ by the same amount is

$$\begin{aligned} & \left. \frac{dG(u(a_1, \omega), \dots, u(a_{m-1}, \omega), u(a_m, \omega) - x, \dots, u(a_n, \omega) - x)}{dx} \right|_{x=0} = - \sum_{i=m}^n \frac{dG(u)}{du(a_i, \omega)} \\ &= \sum_{i=m}^n \int \frac{\beta}{\lambda} p_\beta(a_i | \omega) \left(\sum_{\substack{k=1, \\ k \neq i}}^n p_\beta(a_k | \omega) \left[(v(a_k, \omega) - \beta u(a_k, \omega)) - (v(a_i, \omega) - \beta u(a_i, \omega)) \right] \right) dF(\beta) \\ &= \sum_{i=m}^n \int \frac{\beta}{\lambda} p_\beta(a_i | \omega) \left(\sum_{k=1}^{m-1} p_\beta(a_k | \omega) \left[(v(a_k, \omega) - \beta u(a_k, \omega)) - (v(a_i, \omega) - \beta u(a_i, \omega)) \right] \right) dF(\beta) \\ &\geq \sum_{i=m}^n \int \frac{\beta}{\lambda} p_\beta(a_i | \omega) \left(\sum_{k=1}^{m-1} p_\beta(a_k | \omega) \left[\beta (u(a_i, \omega) - u(a_k, \omega)) - v(a_n, \omega) \right] \right) dF(\beta) \\ &= \sum_{i=m}^n \int \frac{\beta}{\lambda} p_\beta(a_i | \omega) \beta \left(\sum_{k=1}^{m-1} p_\beta(a_k | \omega) \left[u(a_i, \omega) - u(a_k, \omega) - \frac{v(a_n, \omega)}{\beta} \right] \right) dF(\beta), \end{aligned}$$

which is strictly positive since the term in the brackets is strictly positive by (13). Thus, we can decrease $u(a_m, \omega)$ through $u(a_n, \omega)$ until $u(a_i, \omega) \leq u(a_{i-1}, \omega) + \frac{v(a_n, \omega)}{\inf \beta}$, while improving the objective. Thus, each $u(a_i, \omega)$ can be bounded by $(i-1) \frac{v(a_n, \omega)}{\inf \beta}$, which defines a compact set. $\square$

Lemma 4. *There exists an exchangeable utility function $u: A \times \Omega \rightarrow \mathbb{R}$ that maximizes (8) state-by-state.*

Proof. We construct a utility function that maximizes (8) state-by-state and which inherits exchangeability from v . The state-wise objective (8) is symmetric in the components of $v(\cdot, \omega)$. Thus, permuting the components of $v(\cdot, \omega)$ leads to a permutation of the components of the solution candidates $u(\cdot, \omega)$. Define an equivalence relation on Ω : two states are equivalent if their payoff-vectors $v(\cdot, \omega)$ are a permutation of each other. For each equivalence class, pick a representative state and a solution $u(\cdot, \omega)$ of the state-wise objective given that state's payoff-vector. Then, for other states

ω' in the equivalence class, assign the analogous permutation of $u(\cdot, \omega)$, which is a solution given $v(\cdot, \omega')$. This procedure is well-defined because by Assumption 4, for any state ω , all components of $v(\cdot, \omega)$ are pairwise different. Thus, any permutation of $v(\cdot, \omega)$ can be obtained in a *unique* way by a sequence of pairwise permutations of components. The same holds for all components of any state-wise solution $u(\cdot, \omega)$, as by Lemma 2 all components are pairwise different as well. On the equivalence class, any permutation of the state-wise solution of the representative state is thus as likely as any permutation by exchangeability of v . Doing the above procedure for every equivalence class, the resulting u is exchangeable. $\square$

We have shown thus shown that the inner maximization problem separates into state-wise maximization of (8). Next, we show that the overall principal's problem separates by state.

State-Wise Principal's Problem Because the integral over states is a finite sum, we can exchange the order of integration in the implementation cost, which becomes

$$\sum_{\omega \in \Omega} \mu(\omega) \int \left(\sum_{a \in A} p_\beta(a|u(\omega)) \psi(u(a_i, \omega) - v(a_i, \omega)) \right) dF(\beta).$$

Because by constraint u is exchangeable, $p_\beta(a|u(\omega))$ depends only on the state-wise utility vector $u(\omega)$ by (6), so the outer objective separates by state. By the exchangeability assumption on v , one can make a symmetry argument that there is a state-wise minimizer u subject to it being among the state-wise maximizers of (8), such that the actions are exchangeable with respect to u . It follows that if a solution to the principal's problem exists, it must solve for every state $\omega \in \Omega$ the **state-wise principal's problem**:

$$\begin{aligned} \min_{u(\cdot, \omega): A \rightarrow \mathbb{R}} \int \left(\sum_{a \in A} p_\beta(a|u(\omega)) \psi(u(a, \omega) - v(a, \omega)) \right) dF(\beta) \\ \text{s.t.} \\ u(\cdot, \omega) \in \arg \max_{u(\cdot, \omega): A \rightarrow \mathbb{R}} \int \left(-\lambda H(p_\beta(u(\omega))) + \sum_{a \in A} p_\beta(a|u(\omega)) v(a, \omega) \right) dF(\beta) \end{aligned}$$

Lemma 5. *A solution to the state-wise principal's problem exists. The solution is unique if the solution to the inner maximization problem is unique up to a common additive constant.*

Proof. We show again that we can restrict attention to a compact subset of $\mathbb{R}^A$, on which the minimum is obtained by continuity of the objective (ψ is convex implies that ψ is continuous).

First, we reparametrize $(u(a_1, \omega), \dots, u(a_n, \omega))$ by

$$(u_1, \Delta u_2, \dots, \Delta u_n) := (u(a_1, \omega), u(a_2, \omega) - u(a_1, \omega), \dots, u(a_n, \omega) - u(a_1, \omega)).$$

Using $u(a_i, \omega) = u_1 + \Delta u_i$, the outer objective can be rewritten as

$$\sum_{i=1}^n \psi(u_1 + \Delta u_i - v(a_i, \omega)) \int p_\beta(a_i|u(\omega)) dF(\beta) \quad (14)$$

where $p_\beta(a_i|u(\omega))$ depends only on $(\Delta u_2, \dots, \Delta u_n)$. By assumption, ψ is convex and minimized at 0. Thus, keeping $(\Delta u_2, \dots, \Delta u_n)$ fixed, the objective is radially unbounded in u_1 as it is increasing for u_1 to the right of the interval

$$\left[-\max_{i=2, \dots, n} \{\Delta u_i - v(a_i, \omega)\}, -\min_{i=2, \dots, n} \{\Delta u_i - v(a_i, \omega)\} \right]$$

and decreasing when u_1 is to the left of the interval. Hence, because Δu_i is bounded by the proof in Lemma 3, this implies a bound on u_1 . We know that u_1 does not affect the value of the inner objective because it does not affect the choice probabilities, so from the inner maximization problem, there is no other constraint on u_1 . Further, by the proof of Lemma 3, $(\Delta u_2, \dots, \Delta u_n)$ is in a compact set. By Berge's maximum theorem, the set of maximizers form a compact set themselves. Hence, we can restrict attention to a product of two compact sets, which is compact, on which the minimum is obtained.

For the second statement of the Lemma, if the solution to $(\Delta u_2, \dots, \Delta u_n)$ from the inner maximization problem is unique, then the minimization problem above gives a unique solution for u_1 because equation (14) is a strictly convex function in u_1 keeping $(\Delta u_2, \dots, \Delta u_n)$ fixed. $\square$

For $c \in \mathbb{R}$, let $\mathbf{c} := (c, \dots, c) \in \mathbb{R}^A$ (with slight abuse of notation).

Lemma 6. $u(\cdot, \omega) \in \mathbb{R}^A$ solves the state-wise principal's problem given the payoff vector $v(\cdot, \omega) \in \mathbb{R}^A$ if and only if $u(\cdot, \omega) + \mathbf{c}$ does so given payoff vector $v(\cdot, \omega) + \mathbf{c}$.

Proof. As noted in Lemma 3, shifting all components of $v(\cdot, \omega)$ or $u(\cdot, \omega)$ by constants does not affect the inner maximization problem. The solutions to the inner maximization problem given $v(\cdot, \omega)$ and $v(\cdot, \omega) + \mathbf{c}$ are the same as this change only moves the principal's utility by c independent of the agent's choice. And, if $u(\cdot, \omega)$ is a solution to the inner maximization problem, then $u(\cdot, \omega) + \mathbf{c}$ is, too, since shifting all components of $u(\cdot, \omega)$ by a constant does not affect the principal's utility as it does not affect the agent's choice probabilities. Thus, if $u(\cdot, \omega)$ is in the constraint set defined by the inner maximization problem given $v(\cdot, \omega)$, then $u(\cdot, \omega) + \mathbf{c}$ is in the said set given $v(\cdot, \omega) + \mathbf{c}$.

Second, the outer objective is unchanged if $\mathbf{c}$ is added to both $u(\cdot, \omega)$ and $v(\cdot, \omega)$ because the agent's choice probabilities are unaffected and $\psi(u(a_i, \omega) - v(a_i, \omega))$ is unaffected. If $u(\cdot, \omega)$ minimizes the implementation cost given $v(\cdot, \omega)$, then $u(\cdot, \omega) + \mathbf{c}$ minimizes it given $v(\cdot, \omega) + \mathbf{c}$ over the original constraint set plus $\mathbf{c}$. However, this is the actual constraint set, by the above. The other direction follows from shifting by $-\mathbf{c}$. $\square$

Solution to Principal's Problem

Lemma 7. *There exists a solution to the principal's problem such that there is a function R , such that for all $(a, \omega) \in A \times \Omega$,*

$$u(a, \omega) = v(a, \omega) + R(\{v(a, \omega) - v(a', \omega)\}_{a' \in A}).$$

Proof. Define a partition $\mathbb{P}$ on the states based on the equivalence relation that the payoff vector $v(\cdot, \omega) \in \mathbb{R}^A$ of two states are the same up to permutations of its components, shifting all components by a constant, and combinations of both. Formally, $\omega \sim \omega'$ if there exists a permutation $\tilde{v}(\cdot, \omega)$ of $v(\cdot, \omega)$ and a $c \in \mathbb{R}$ such that $v(\cdot, \omega) = \tilde{v}(\cdot, \omega) + \mathbf{c}$. We proceed by constructing $u(\cdot, \omega)$ on each partition element.

Let $P \in \mathbb{P}$ be some partition element and ω be some state in P . Define $u(\cdot, \omega)$ to be a solution to the state-wise principal's problem, which exists by Lemma 5. So, for each other state ω' in P , there exist a $c \in \mathbb{R}$ and a permutation $\tilde{v}(\cdot, \omega)$ of the payoff vector $v(\cdot, \omega)$, such that $v(\cdot, \omega') = \mathbf{c} + \tilde{v}$. Moreover, the constant c and the way $v(\cdot, \omega)$ is permuted are unique, the latter because by Assumption 4. Define $u(\cdot, \omega')$ to be the analogous permutation of $u(\cdot, \omega)$ plus $\mathbf{c}$. This gives a solution to the state-wise principal's problem under ω' because the state-wise principal's problem is the same up to this permutation and by Lemma 6. Because the actions are exchangeable with respect to $v(\cdot, \omega)$, each state in partition element in P that has $c = 0$ relative to ω (that is a “pure permutation”) has the same probability. Hence, by construction $u(\cdot, \omega)$ is a pure permutation on these elements, so the actions are exchangeable with respect to $u(\cdot, \omega)$. Thus, the constructed $u(\cdot, \omega)$ solves the principal's problem.

Further, $u(\cdot, \omega)$ is a function only of the payoff vector $v(\cdot, \omega)$. Denote this function by $M: \mathbb{R}^A \rightarrow \mathbb{R}^n$ and by M_i the i -th component of M . Define for all $(v_1, \dots, v_{n-1}) \in \mathbb{R}^A$

$$\tilde{R}(v_1, \dots, v_{n-1}) := M_n(v_1, \dots, v_{n-1}, 0).$$

By construction, a permutation of the arguments of M permutes the image in an analogous way. Formally, for all pairwise different $i, j, k \in \{1, \dots, n\}$

$$\begin{aligned} M_i(v_1, \dots, v_i, \dots, v_j, \dots, v_n) &= M_j(v_1, \dots, v_j, \dots, v_i, \dots, v_n) \\ M_k(v_1, \dots, v_i, \dots, v_j, \dots, v_n) &= M_k(v_1, \dots, v_j, \dots, v_i, \dots, v_n) \end{aligned}$$

This implies that $\tilde{R}$ is invariant under relabeling of its components, so we can define

$$R(\{v_1, \dots, v_n\}) := \tilde{R}(-v_1, \dots, -v_n).$$

We have by Lemma 6,

$$\begin{aligned}
u(a_i, \omega) &= M_i(v(a_1, \omega), \dots, v(a_2, \omega)) \\
&= v(a_i, \omega) + M_i(v(a_1, \omega) - v(a_i, \omega), \dots, v(a_i, \omega) - v(a_i, \omega), \dots, v(a_2, \omega) - v(a_i, \omega)) \\
&= v(a_i, \omega) + M_n(v(a_1, \omega) - v(a_i, \omega), \dots, v(a_2, \omega) - v(a_i, \omega), 0) \\
&= v(a_i, \omega) + \tilde{R}(v(a_1, \omega) - v(a_i, \omega), \dots, v(a_2, \omega) - v(a_i, \omega)) \\
&= v(a_i, \omega) + R(\{v(a_i, \omega) - v(a', \omega)\}_{a' \in A \setminus \{a_i\}})
\end{aligned}$$

□

Note that the constructed solution to the state-wise principal's problem depends only on the state-wise actual and counterfactual payoffs as well as the distribution over β and λ . Thus, the solution does not depend on the whole on the decision-problem that the principal delegates.

C Theorem 1

Proof. By Theorem 2 there is a solution with

$$u(a, \omega) = v(a, \omega) + R(v(a, \omega) - v(a_-, \omega)), \quad (15)$$

if $v(a, \omega) - v(a_-, \omega) \neq 0$ for all states (because we have assumed strict preferences of the principal for that theorem). First, we show that this statement extends to cases where $v(a, \omega) - v(a_-, \omega) = 0$ for some states. Next, we show that any two such solutions have R and Q that are unique up to a measure zero set in $\mathbb{R}$ (with respect to the Lebesgue measure). Then, we show that any such solution satisfies Properties 1 and 2.

Extending Theorem 1 By Theorem 2, a solution to the principal's problem exists and every solution must, in every state, solve the state-wise principal's problem, which is

$$\begin{aligned}
&\min_{u(\cdot, \omega): A \rightarrow \mathbb{R}} \int \left(\sum_{i=1}^2 p_\beta(a_i | u(\omega)) \psi(u(a_i, \omega) - v(a_i, \omega)) \right) dF(\beta) \\
&\quad \text{s.t.} \\
&u(\cdot, \omega) \in \arg \max_{u(\cdot, \omega): A \rightarrow \mathbb{R}} \int \left(\lambda H(p_\beta(u(\omega))) + \sum_{i=1}^2 p_\beta(a_i | u(\omega)) v(a_i, \omega) \right) dF(\beta)
\end{aligned}$$

If $v(a_1, \omega) = v(a_2, \omega)$, the payoff $\sum_{i=1}^2 p_\beta(a_i | u(\omega)) v(a_i, \omega)$ is unaffected by the conditional choice probabilities $p_\beta(a_i | u(\omega))$. The entropy term is maximized if

$$p_\beta(a_1 | u(\omega)) = p_\beta(a_2 | u(\omega)) = 1/2,$$

which is achieved for all types if and only if $u(a_1, \omega) = u(a_2, \omega)$. In that case, because the implementation cost is minimized at 0, the unique solution is

$$u(a_1, \omega) = u(a_2, \omega) = v(a_1, \omega) = v(a_2, \omega).$$

Take our solution from Theorem 2 for states where $v(a_1, \omega) \neq v(a_2, \omega)$, we can extend it for states where $v(a_1, \omega) = v(a_2, \omega)$ and maintain exchangeability, so it must be a solution.

Uniqueness By (9), the inner objective can be rewritten as

$$\int \left(\sum_{i=1}^2 p_\beta(a_i|u(\omega)) (v(a_i, \omega) - \beta u(a_i, \omega)) + \lambda \log \left(\sum_{i=1}^2 e^{\frac{\beta}{\lambda} u(a_i, \omega)} \right) \right) dF(\beta).$$

Using $p_\beta(a_1|\omega) = 1 - p_\beta(a_2|\omega)$, we can rewrite the objective as

$$\begin{aligned} \int p_\beta(a_2|u(\omega)) \left[v(a_2, \omega) - v(a_1, \omega) - \beta(u(a_2, \omega) - u(a_1, \omega)) \right] + v(a_1, \omega) - \beta u(a_1, \omega) \\ + \lambda \left(\frac{\beta}{\lambda} u(a_1, \omega) + \log \left(1 + e^{\frac{\beta}{\lambda} (u(a_2, \omega) - u(a_1, \omega))} \right) \right) dF(\beta) \end{aligned}$$

Defining $\Delta u(\omega) = u(a_2, \omega) - u(a_1, \omega)$ and $\Delta v(\omega) = v(a_2, \omega) - v(a_1, \omega)$, and subtracting the constant $v(a_1, \omega)$, we obtain that the objective is

$$\int \left(p_\beta(a_2|u(\omega)) (\Delta v(\omega) - \beta \Delta u(\omega)) + \lambda \log \left(1 + e^{\frac{\beta}{\lambda} \Delta u(\omega)} \right) \right) dF(\beta) \quad (16)$$

Note that $p_\beta(a_2|u(\omega))$ depends only on $\Delta u(\omega)$ by (6), so with slight abuse of notation, we write $p_\beta(a_2|\Delta u(\omega))$. The first-order condition is

$$\int \left(\frac{dp_\beta(a_2|\Delta u(\omega))}{d\Delta u(\omega)} (\Delta v(\omega) - \beta \Delta u(\omega)) \right) dF(\beta) = 0, \quad (17)$$

which can be rewritten as

$$\Delta u(\omega) = \Delta v(\omega) \left(\frac{\int \frac{dp_\beta(a_2|\Delta u(\omega))}{d\Delta u(\omega)} \beta dF(\beta)}{\int \frac{dp_\beta(a_2|\Delta u(\omega))}{d\Delta u(\omega)} dF(\beta)} \right)^{-1}. \quad (18)$$

as used in the main text.

The solution $\Delta u(\omega)$ need not be unique. However, using the supermodularity of the objective in $\Delta u(\omega)$ and $\Delta v(\omega)$, we show that there is a unique maximizer $\Delta u(\omega)$ for almost all $\Delta v(\omega)$.

Lemma 8. *The set of solutions $\Delta u(\omega)$ as a function of $\Delta v(\omega)$ are increasing in the strong set order. For almost all $\Delta v(\omega)$, the solution $\Delta u(\omega)$ is unique.*

Proof. The objective (16) has a positive cross-derivative in $\Delta v(\omega)$ and $\Delta u(\omega)$, which, using (17), is

$$\int \frac{dp_\beta(a_2|\Delta u(\omega))}{d\Delta u(\omega)} dF(\beta) > 0.$$

It is positive because $p_\beta(a_2|\Delta u(\omega))$ is strictly increasing in $\Delta u(\omega)$ for all $\beta > 0$. Thus, by Topkis characterization theorem (Topkis, 1978), the objective is supermodular in $\Delta u(\omega)$ and $\Delta v(\omega)$. By Topkis theorem (ibid.), the set of $\Delta u(\omega)$ that maximize the objective as a function of $\Delta v(\omega)$ is increasing in the strong set order. Since the cross-partial is strictly positive, an even stronger statement holds: if $\Delta u'' > \Delta u'$ are both maximizers at $\Delta v(\omega)$, then at any $\Delta v(\omega)' > \Delta v(\omega)$, $\Delta u''$ achieves a strictly higher payoff than $\Delta u'$. That is because the derivative of the objective between $\Delta u'$ and $\Delta u''$ is strictly higher at any point, so the integral over the derivative is now positive, where it was 0 before. Thus, any selection function of maximizers as a function of $\Delta v(\omega)$ must have a jump discontinuity wherever the set of maximizers is not a singleton. Since monotonic functions can have at most countably many jump discontinuities, there are at most countably many $\Delta v(\omega)$ for which there is not a unique maximizer. Thus, for almost all $\Delta v(\omega)$, the solution is unique. $\square$

By Lemma 5 and 8, the solution to the outer minimization problem subject to the inner maximization problem is unique for almost all $\Delta v(\omega)$. Thus, $R(x)$ is unique for almost all $x \in \mathbb{R}$ and thereby also $Q(x) := x + R(x) - R(-x)$, which equals $\Delta u(\omega)$ given $\Delta v(\omega) = x$.

Property 1 Above, we have already argued that $R(0) = 0$. First, we show that $R(x) \geq 0$ for $x > 0$. Note that equation (18) show that $Q(x) - x > 0$ when $x > 0$. Thus, if $R(x)$ was negative then $R(x)$ and $R(-x)$ were negative by $R(x) - R(-x) = Q(x) - x$. Then, we could increase $R(x)$ and $R(-x)$ by the same small amount (thus maintaining $Q(x)$ being a solution to the state-wise maximization problem) and decrease the implementation cost by ψ being strictly convex and minimal at 0. Analogously, one can show that $R(x) \leq 0$ if $x < 0$.

If $R(x)$ was 0, then $R(-x)$ would be negative by $R(x) - R(-x) = Q(x) - x$. Because ψ is differentiable and has a global minimum at 0, it must have derivative 0 at 0. Then, we could increase $R(x)$ and $R(-x)$ by the some small enough amount (thus maintaining $Q(x)$ being a solution to the state-wise maximization problem) and decrease the implementation cost, by ψ being strictly convex and having derivative 0 at 0. Analogously, one can show that $R(x) < 0$ if $x < 0$.

Property 2 As argued in the main text, Property 2 is equivalent to $Q(x)/x$ being strictly increasing on $\mathbb{R}_{>0}$. We show that any $Q(x) := x + R(x) - R(-x)$ satisfies this property. Rearranging (18), we get

$$\frac{Q(x)}{x} = \left(\frac{\int \frac{dp_\beta(a_2|Q(x))}{dQ(x)} \beta dF(\beta)}{\int \frac{dp_\beta(a_2|Q(x))}{dQ(x)} dF(\beta)} \right)^{-1}. \quad (19)$$

Because β has a discrete distribution by Assumption 1, the integral $\int g(\beta) dF(\beta)$ over some measurable function $g(\beta)$ equals $\sum_{\beta \in \text{supp}(\beta)} g(\beta) f(\beta)$ with probability mass function f . The term in

parentheses in (19) is the expectation of β with respect to the probability mass function

$$\frac{\frac{dp_\beta(a_2|Q(x))}{dQ(x)} f(\beta)}{\int \frac{dp_b(a_2|Q(x))}{dQ(x)} dF(b)}.$$

Since we are considering $x > 0$, we have $Q(x) > 0$. Lemma 9 below shows that this probability mass function undergoes a strict likelihood-ratio shift downwards as $Q(x)$ increases for $Q(x) > 0$. Because $Q(x)$ is strictly increasing by Lemma 8 and because strict likelihood-ratio dominance implies strict first-order stochastic dominance, this implies that the term in parentheses in (19) is strictly decreasing in x . Thus, $Q(x)/x$ is strictly increasing in x .

Lemma 9. *The probability mass function*

$$\frac{\frac{dp_\beta(a_2|Q(x))}{dQ(x)} f(\beta)}{\int \frac{dp_b(a_2|Q(x))}{dQ(x)} dF(b)}$$

undergoes a strict likelihood-ratio shift downwards as $Q(x)$ increases for $Q(x) > 0$.

Proof. We need to show that for all $1 \geq \alpha > \beta > 0$,

$$\frac{\left(\frac{\frac{dp_\alpha(a_2|Q(x))}{dQ(x)} f(\alpha)}{\int \frac{dp_b(a_2|Q(x))}{dQ(x)} dF(b)} \right)}{\left(\frac{\frac{dp_\beta(a_2|Q(x))}{dQ(x)} f(\beta)}{\int \frac{dp_b(a_2|Q(x))}{dQ(x)} dF(b)} \right)}$$

is decreasing in $Q(x)$. Canceling out the integration constants in the denominators, this term is equal to

$$\frac{\frac{dp_\alpha(a_2|Q(x))}{dQ(x)} f(\alpha)}{\frac{dp_\beta(a_2|Q(x))}{dQ(x)} f(\beta)} = \frac{\left(\frac{\frac{\alpha}{\lambda} e^{\frac{\alpha}{\lambda} Q(x)} f(\alpha)}{\left(1 + e^{\frac{\alpha}{\lambda} Q(x)}\right)^2} \right)}{\left(\frac{\frac{\beta}{\lambda} e^{\frac{\beta}{\lambda} Q(x)} f(\beta)}{\left(1 + e^{\frac{\beta}{\lambda} Q(x)}\right)^2} \right)} = \frac{\alpha f(\alpha)}{\beta f(\beta)} e^{\frac{\alpha-\beta}{\lambda} Q(x)} \left(\frac{1 + e^{\frac{\beta}{\lambda} Q(x)}}{1 + e^{\frac{\alpha}{\lambda} Q(x)}} \right)^2$$

Ignoring the constant $\frac{\beta f(\beta)}{\alpha f(\alpha)}$, the term is decreasing in $Q(x)$ if the logarithm of the term is decreasing in $Q(x)$.

$$\log \left(e^{\frac{\alpha-\beta}{\lambda} Q(x)} \left(\frac{1 + e^{\frac{\beta}{\lambda} Q(x)}}{1 + e^{\frac{\alpha}{\lambda} Q(x)}} \right)^2 \right) = (\alpha - \beta) \frac{Q(x)}{\lambda} + 2 \log \frac{1 + e^{\frac{\beta}{\lambda} Q(x)}}{1 + e^{\frac{\alpha}{\lambda} Q(x)}}$$

Differentiating with respect to $Q(x)$:

$$\begin{aligned}
\frac{d}{dQ(x)} \left((\alpha - \beta) \frac{Q(x)}{\lambda} + 2 \log \frac{1 + e^{\frac{\beta}{\lambda} Q(x)}}{1 + e^{\alpha Q(x)/\lambda}} \right) &= \frac{\alpha - \beta}{\lambda} + 2 \frac{\frac{d}{dQ(x)} \frac{1 + e^{\frac{\beta}{\lambda} Q(x)}}{1 + e^{\alpha Q(x)/\lambda}}}{\frac{1 + e^{\frac{\beta}{\lambda} Q(x)}}{1 + e^{\alpha Q(x)/\lambda}}} \\
&= \frac{\alpha - \beta}{\lambda} + 2 \frac{\frac{\beta}{\lambda} e^{\frac{\beta}{\lambda} Q(x)} (1 + e^{\alpha Q(x)/\lambda}) - \frac{\alpha}{\lambda} e^{\alpha Q(x)/\lambda} (1 + e^{\frac{\beta}{\lambda} Q(x)})}{(1 + e^{\alpha Q(x)/\lambda})^2} \frac{1 + e^{\alpha Q(x)/\lambda}}{1 + e^{\frac{\beta}{\lambda} Q(x)}} \\
&= \frac{\alpha - \beta}{\lambda} + 2 \frac{\beta - \alpha}{\lambda} \frac{e^{\frac{\alpha + \beta}{\lambda} Q(x)}}{(1 + e^{\frac{\beta}{\lambda} Q(x)})(1 + e^{\alpha Q(x)/\lambda})} + 2 \frac{\frac{\beta}{\lambda} e^{\frac{\beta}{\lambda} Q(x)} - \frac{\alpha}{\lambda} e^{\alpha Q(x)/\lambda}}{(1 + e^{\frac{\beta}{\lambda} Q(x)})(1 + e^{\alpha Q(x)/\lambda})}
\end{aligned}$$

Multiply by $\lambda(1 + e^{\frac{\beta}{\lambda} Q(x)})(1 + e^{\alpha Q(x)/\lambda}) > 0$ gives

$$\begin{aligned}
&(\alpha - \beta)(1 + e^{\frac{\beta}{\lambda} Q(x)})(1 + e^{\alpha Q(x)/\lambda}) + 2(\beta - \alpha)e^{\frac{\alpha + \beta}{\lambda} Q(x)} + 2(\beta e^{\frac{\beta}{\lambda} Q(x)} - \alpha e^{\alpha Q(x)/\lambda}) \\
&= (\alpha - \beta) + (\alpha + \beta)e^{\frac{\beta}{\lambda} Q(x)} - (\alpha + \beta)e^{\alpha Q(x)/\lambda} + (\beta - \alpha)e^{\frac{\alpha + \beta}{\lambda} Q(x)} \\
&= (\alpha + \beta)(e^{\frac{\beta}{\lambda} Q(x)} - e^{\alpha Q(x)/\lambda}) + (\alpha - \beta)(1 - e^{\frac{\beta}{\lambda} Q(x)}e^{\alpha Q(x)/\lambda}).
\end{aligned}$$

By $\alpha, \beta > 0$ and $Q(x) > 0$, we have $e^{\frac{\alpha}{\lambda} Q(x)} > e^{\frac{\beta}{\lambda} Q(x)}$ and hence the first term is negative. By $\alpha > \beta$ and $Q(x) > 0$, we have $e^{\frac{\beta}{\lambda} Q(x)} > 1$, $e^{\alpha Q(x)/\lambda} > 1$, and $e^{\frac{\beta}{\lambda} Q(x)}e^{\alpha Q(x)/\lambda} > 1$. Hence, the second term is negative. This establishes the negative derivative in $Q(x)$. $\square$

Thus, Q and R of the regret solution satisfy

$$\begin{aligned}
Q(x) &\in \arg \max_{y \in \mathbb{R}} \int \frac{e^{\beta y/\lambda}}{1 + e^{\beta y/\lambda}} (x - \beta y) + \lambda \log(1 + e^{\beta y/\lambda}) dF(\beta) \\
R(x) &= \arg \min_{R \in \mathbb{R}} \left(\int p_\beta(Q(x)) dF(\beta) \right) \psi(R) + \left(1 - \int p_\beta(Q(x)) dF(\beta) \right) \psi(x + R - Q(x))
\end{aligned}$$

where $p_\beta(Q(x)) := \frac{e^{\frac{\beta}{\lambda} Q(x)}}{1 + e^{\frac{\beta}{\lambda} Q(x)}}$. $\square$

D Proposition 1

Proof. Let preferences have an original regret representation. It is easy to see that for nonnegative functions Q with $Q(0) = 0$, strict convexity on $[0, \infty)$ implies strict starshapedness. Therefore, preferences have an evolutionary regret representation.

Let preferences have an evolutionary regret representation. Define $x(a, \omega) := v(a, \omega)$ and $\Psi(x, y) := Q(x - y)$. Then, preferences are represented by (3). By skew-symmetry of Q , Ψ is skew-symmetric. It remains to show that Ψ satisfies ordering, increasingness, and convexity. First, note that since Q is skew-symmetric, $Q(0) = 0$. And since Q is starshaped, Q is strictly increasing.

Ordering: Let $x, y \in \mathbb{R}$. Since $Q(0) = 0$ and Q is strictly increasing, we have $x - y \geq 0 \Leftrightarrow Q(x - y) \geq 0$. By $\Psi(x, y) = Q(x - y)$, we obtain $x \geq y \Leftrightarrow \Psi(x, y) \geq 0$.

Increasingness: Let $x, y, z \in \mathbb{R}$. First, $x > y$ is equivalent to $x - z > y - z$, which is equivalent to $Q(x - y) > Q(y - z)$ by Q being strictly increasing. This is equivalent to $\Psi(x, y) > \Psi(y, z)$ by definition of Ψ . Second, since $Q(0) = 0$ and Q is strictly increasing, $x = y$ is equivalent to $Q(x - y) = 0$, which is equivalent to $\Psi(x, y) = 0$.

Convexity: Let $x, y, z \in \mathbb{R}$ with $x > y > z$. Because Q is starshaped with $Q(0) = 0$, Q is strictly superadditive. A “non-strict” version of this statement is shown, for example, in Bruckner and Ostrow (1962), Theorem 5, and a strict version follows by the same proof. Thus, $Q(x - z) > Q(x - y) + Q(y - z)$ and therefore $\Psi(x, y) > \Psi(x, y) + \Psi(y, z)$. $\square$

E Proposition 2

Proof. Loomes and Sugden (1982) show that the Allais paradox and the common-ratio effect are equivalent to strict superadditivity,

$$\forall x_1, x_2: 0 < x_2 < x_1 \rightarrow Q(x_1) - Q(x_1 - x_2) - Q(x_2) > 0, \quad (20)$$

and that simultaneous gambling and insurance is equivalent to

$$\forall x > 0: Q\left(\frac{p}{1-p}x\right) \geq \frac{p}{1-p}Q(x) \quad \text{if } p \geq 0.5. \quad (21)$$

We show that (20) and (21) are implied by $Q(x)/x$ being strictly increasing on $\mathbb{R}_{>0}$ and that (21) implies that $Q(x)/x$ is strictly on $\mathbb{R}_{>0}$.

To show that (20) is implied by Property 2, let $x_1 > x_2 > 0$ and define $Q_1 := Q(x_1)$, $Q_2 := Q(x_2)$, and $Q_{12} := Q(x_1 - x_2)$. We can assume without loss $0 < x_1 - x_2 < x_2 < x_1$. That is because if $0 < x_2 < x_1 - x_2 < x_1$, we can simply exchange x_2 and $x_1 - x_2$ in (20). By $Q(x)/x$ being strictly increasing on $\mathbb{R}_{>0}$:

$$\begin{aligned} \frac{Q_1}{x_1} &> \frac{Q_2}{x_2} \\ \Rightarrow Q_1 x_2 &> Q_2 x_1 \\ \Rightarrow Q_1 x_2 - Q_2 x_2 &> Q_2 x_1 - Q_2 x_2 \\ \Rightarrow (Q_1 - Q_2) x_2 &> Q_2 (x_1 - x_2) \\ \Rightarrow \frac{Q_1 - Q_2}{x_1 - x_2} &> \frac{Q_2}{x_2} \end{aligned}$$

Using again that $Q(x)/x$ is strictly increasing on $\mathbb{R}_{>0}$,

$$\frac{Q_2}{x_2} > \frac{Q_{12}}{x_1 - x_2}.$$

Together,

$$\frac{Q_1 - Q_2}{x_1 - x_2} > \frac{Q_2}{x_2} > \frac{Q_{12}}{x_1 - x_2} \Rightarrow Q_1 - Q_{12} - Q_2 > 0.$$

Equation (21) follows from $Q(x)/x$ strictly increasing on $\mathbb{R}_{>0}$ by

$$p > 0.5 \Rightarrow \frac{p}{1-p} > 1 \Rightarrow \frac{Q(\frac{p}{1-p}x)}{\frac{p}{1-p}x} > \frac{Q(x)}{x} \Rightarrow Q(\frac{p}{1-p}x) > \frac{p}{1-p}Q(x),$$

and analogously for $p < 0.5$. When $p = 0.5$, $\frac{p}{1-p} = 1$ and the statement trivially holds.

For the other direction, suppose $Q(x)/x$ was not strictly increasing on $\mathbb{R}_{>0}$. Then, there exists $0 < x < y$ such that $\frac{Q(y)}{y} \leq \frac{Q(x)}{x}$. Let $p = \frac{y/x}{1+y/x} \in (1/2, 1)$. Then, $\frac{p}{1-p} = \frac{y}{x}$ and thus

$$\frac{Q(\frac{p}{1-p}x)}{\frac{p}{1-p}x} \leq \frac{Q(x)}{x} \Rightarrow Q(\frac{p}{1-p}x) \leq \frac{p}{1-p}Q(x)$$

despite $p > 1/2$ violating (21). $\square$

F Corollary 1

Proof. Regarding the action-implementation stage, we need to establish that given the information the agent of type β acquires, he takes the principal's preferred action. That is, we show that the posterior of type β under some action $a_i \in A$ makes a_i the optimal choice for the principal. This posterior is proportional to the product of the prior and the likelihood, the latter of which is given by the state-dependent choice probabilities. Consider the posterior probability of some emotion-exclusive payoff vector $(v(a_i, \omega), v(a_{-i}, \omega)) = (v_1, v_2)$ and its permutation (v_2, v_1) with $v_1 > v_2$. Both payoff vectors have the same prior probability by exchangeability. Because emotions preserve the state-wise ordering of payoffs, the conditional choice probability of any agent β of action a_i is higher than the one of a_{-i} by (6). Thus, the posterior probability of (v_1, v_2) conditional on a_i is higher than conditional on a_{-i} , contributing to a higher expected utility for the principal of a_i than a_{-i} . Integrating over all pairs of payoff vectors and their permutation, the expected utility of action a_i with respect to v is higher than the one of action a_{-i} . $\square$

G Corollary 2

Proof. Let $x > 0$ and denote $p_\beta(Q(x)) := \frac{e^{\frac{\beta}{\lambda}Q(x)}}{1 + e^{\frac{\beta}{\lambda}Q(x)}}$. From the proof of Theorem 1, regret and rejoicing, $R(-x)$ and $R(x)$, minimize the implementation cost subject to being consistent with $Q(x)$:

$$\begin{aligned} \min_{R(x), R(-x) \in \mathbb{R}} & \left(\int p_\beta(Q(x)) dF(\beta) \right) \psi(R(x)) + \left(1 - \int p_\beta(Q(x)) dF(\beta) \right) \psi(R(-x)) \\ & \text{s.t.} \\ & x + R(x) - R(-x) = Q(x) \end{aligned}$$

That x is positive implies $Q(x) > 0$, which in turn implies that for all $\beta > 0$:

$$\begin{aligned} p_\beta(Q(x)) &= \frac{e^{\frac{\beta}{\lambda}Q(x)}}{1 + e^{\frac{\beta}{\lambda}Q(x)}} > \frac{1}{2} > 1 - p_\beta(Q(x)) \\ \Rightarrow \int p_\beta(Q(x))dF(\beta) &> 1 - \int p_\beta(Q(x))dF(\beta) \end{aligned}$$

Thus, in the implementation-cost minimization problem, the weight on $\psi(R(x))$ is greater than the weight on $\psi(R(-x))$. Suppose for the sake of contradiction that we had $R(x) > |R(-x)|$. By convexity of ψ and symmetry around 0, there is an $\varepsilon > 0$ such that the implementation cost decreases if $R(x)$ and $R(-x)$ are both lowered by ε , which maintains the constraint $x + R(x) - R(-x) = Q(x)$. $\square$

H Corollary 3

Proof. We show, first, that decreasing λ increases $Q(x)$ for all $x > 0$ and, second, that smaller λ and greater $Q(x)$ lead to a greater $R(-x)$.

First, decreasing λ has the same effect as increasing $Q(x)$ in Lemma 9 because both enter the density in (18) only through $\frac{Q(x)}{\lambda}$. Thus, decreasing λ leads to a likelihood-ratio shift downwards and by (18) to a greater $Q(x)$ for any $x > 0$.

Second, given $Q(x)$, $R(-x)$ is determined by the implementation-cost minimization problem. Inserting the constraint $x + R(x) + R(-x) = Q(x)$, we can rewrite the implementation cost Ψ with slight abuse of notation as a function of $R(-x)$, $Q(x)$, and λ :

$$\begin{aligned} \Psi(R(-x), Q(x), \lambda) &:= \left(\int \frac{e^{\frac{\beta}{\lambda}Q(x)}}{1 + e^{\frac{\beta}{\lambda}Q(x)}} dF(\beta) \right) \psi(Q(x) - x + R(-x)) + \\ &\quad \left(1 - \int \frac{e^{\frac{\beta}{\lambda}Q(x)}}{1 + e^{\frac{\beta}{\lambda}Q(x)}} dF(\beta) \right) \psi(R(-x)) \end{aligned}$$

We show that the Ψ satisfies strictly increasing differences in $(R(-x), (Q(x), -\lambda))$, which implies that the solution $R(-x)$ to the minimization problem is strictly smaller, when $Q(x)$ is greater and λ is smaller by Topkis Theorem. By Topkis Characterization Theorem, it is sufficient to show that $\frac{d^2\Psi}{dQ(x)dR(-x)} > 0$ and $-\frac{d^2\Psi}{d\lambda dR(-x)} = \frac{d^2\Psi}{d(-\lambda)dR(-x)} > 0$. We can restrict attention to $Q(x) > 0$, $R(-x) < 0$ and $R(x) = Q(x) - x + R(-x) > 0$, which are satisfied at the optimum by Theorem 1 anyways.

$$\begin{aligned} \frac{d\Psi}{dR(-x)} &= \left(\int \frac{e^{\frac{\beta}{\lambda}Q(x)}}{1 + e^{\frac{\beta}{\lambda}Q(x)}} dF(\beta) \right) \psi'(Q(x) - x + R(-x)) \\ &\quad + \left(1 - \int \frac{e^{\frac{\beta}{\lambda}Q(x)}}{1 + e^{\frac{\beta}{\lambda}Q(x)}} dF(\beta) \right) \psi'(R(-x)) \end{aligned}$$

$$\begin{aligned} \Rightarrow \frac{d^2\Psi}{dQ(x)dR(-x)} &= \left(\int \frac{d}{dQ(x)} \frac{e^{\frac{\beta}{\lambda}Q(x)}}{1 + e^{\frac{\beta}{\lambda}Q(x)}} dF(\beta) \right) \psi'(Q(x) - x + R(-x)) \\ &\quad + \left(\int \frac{e^{\frac{\beta}{\lambda}Q(x)}}{1 + e^{\frac{\beta}{\lambda}Q(x)}} dF(\beta) \right) \psi''(Q(x) - x + R(-x)) \\ &\quad - \left(\int \frac{d}{dQ(x)} \frac{e^{\frac{\beta}{\lambda}Q(x)}}{1 + e^{\frac{\beta}{\lambda}Q(x)}} dF(\beta) \right) \psi'(R(-x)) \end{aligned}$$

By differentiability of the integrand in Q and boundedness of this derivative, we exchanged integration over β and differentiation by $Q(x)$ according to Leibniz rule. Since ψ is strictly convex and minimal at 0, we have $\psi'(x) \geq 0 \Leftrightarrow x \geq 0$ and $\psi''(x) > 0$. By the constraints on $Q(x)$ and $R(-x)$, we have $R(-x) < 0$ and $Q(x) - x + R(-x) > 0$, so $\psi'(R(-x)) < 0$ and $\psi'(Q(x) - x + R(-x)) > 0$. Finally, by

$$\begin{aligned} \frac{e^{\frac{\beta}{\lambda}Q(x)}}{1 + e^{\frac{\beta}{\lambda}Q(x)}} &> 0 \quad \text{and} \\ \frac{d}{dQ(x)} \frac{e^{\frac{\beta}{\lambda}Q(x)}}{1 + e^{\frac{\beta}{\lambda}Q(x)}} &> 0 \end{aligned}$$

for $Q(x) > 0$, we obtain that $\frac{d^2\Psi}{dQ(x)dR(-x)}$ is strictly positive.

Regarding λ ,

$$\begin{aligned} \frac{d^2\Psi}{d\lambda dR(-x)} &= \left(\int \frac{d}{d\lambda} \frac{e^{\frac{\beta}{\lambda}Q(x)}}{1 + e^{\frac{\beta}{\lambda}Q(x)}} dF(\beta) \right) \psi'(Q(x) - x + R(-x)) \\ &\quad - \left(\int \frac{d}{d\lambda} \frac{e^{\frac{\beta}{\lambda}Q(x)}}{1 + e^{\frac{\beta}{\lambda}Q(x)}} dF(\beta) \right) \psi'(R(-x)) \end{aligned}$$

By

$$\frac{d}{d\lambda} \frac{e^{\frac{\beta}{\lambda}Q(x)}}{1 + e^{\frac{\beta}{\lambda}Q(x)}} < 0$$

and we obtain that $-\frac{d^2\Psi}{d\lambda dR(-x)} < 0$ and hence $\frac{d^2\Psi}{d(-\lambda)dR(-x)} > 0$. □

VI Appendix B: Incomplete Feedback

A Incomplete Feedback

So far, we have assumed that the principal can choose emotions as an arbitrary function of the action and realized state. In many decision contexts, however, ex post only the payoff is revealed but not the full state. For example, if the agent pursues some economic project, the realized return might be observable but not the return had he pursued another project. In particular, when only

the realized payoff $v(a, \omega)$ is learned but not the state ω , the principal cannot in general infer the payoffs of counterfactual actions. Thus, she cannot implement the general regret solution from the previous section. This section generalizes the principal's problem to account for incomplete feedback under the simplifying assumption of a known bias.

As noted earlier, under known bias it is possible to implement the first-best attention strategy by offering a contract that scales up the payoff-differences by $1/\beta$. However, to find the emotion scheme m with minimal variance, we need to determine *all* emotion schemes that implement the optimal attention strategy. Below, we argue how this exercise teaches us something about the nature of the optimal mechanism when the first-best attention strategy cannot be implemented due to an uncertain bias.

Ex-post feedback also plays an important role in the study of regret and disappointment. While classical regret theory (Bell, 1982; Loomes and Sugden, 1982) does not take into account feedback, several experiments have documented that regret affects decisions more if feedback is expected (Zeelenberg, 1999a). To accommodate for that evidence, Humphrey (2004) introduces a theory of feedback-conditional regret which employs two regret-rejoicing functions, one for complete feedback and one for incomplete feedback. Gabillon (2020) proposes a regret theory for arbitrary feedback structures, where the regret-rejoicing function is applied to the difference between the actual payoff and the highest counterfactual payoff, given the posterior beliefs after feedback. Evidence from Camille et al. (2004) based on self-reported emotions, skin conductance responses, and choices suggests, however, that under partial feedback (only the payoff of the chosen action is learned) primarily disappointment and elation are experienced, while under complete feedback mainly regret and rejoicing are experienced. We are not aware of theories that combine regret and disappointment depending on the feedback structure.

To incorporate feedback into our model, we assume that the principal can choose emotions only as a function of the feedback that is received ex post. Formally, we assume the emotion scheme m is measurable with respect to a feedback structure $\mathcal{F}: A \times \Omega \rightarrow Z$ where Z is a message space, that is,

$$\forall a, a' \in A, \omega, \omega' \in \Omega: \mathcal{F}(a, \omega) = \mathcal{F}(a', \omega') \rightarrow m(a, \omega) = m(a', \omega').$$

We consider partitional feedback structures that reveal the payoff $v(a, \omega)$, the action a , and a partition $\mathcal{P}$ of the state space.²⁰ In one extreme case, the feedback structure reveals the state fully (the finest partition; complete feedback) and in the other extreme, only the payoff is revealed (the coarsest partition; partial feedback). In intermediate cases, some information about the state is revealed, such as general market conditions, while more specific information, such as how each particular economic project would have done, is not revealed.

²⁰A partition of the state space is a set of mutually exclusive and exhaustive subsets $\mathcal{P} = \{P_1, \dots, P_n\}$ of Ω . In the Appendix, we show that the resulting emotions are essentially of the same form, when we allow for other deterministic feedback structures that reveal at least the payoff. Further, none of the results below would change if the action was not revealed.

In some decision-problems, however, the agent can back out more information about the state from the material payoff and the action. To rule out such cases, we make an additional assumption that the message space is a “connected” on each partition. To formulate this condition, we endow the message space, Z , with a symmetric relation $\sim$, which makes Z an undirected graph where $\sim$ stands for the edges of the graph. We define $z \sim z'$ if and only if there exists $\omega \in \Omega$ and $a, a' \in A$ such that $z = \mathcal{F}(a, \omega)$ and $z' = \mathcal{F}(a', \omega)$. Thus, two messages are directly connected by an edge if they are observed under the same state. Because the state-wise payoff-differences determine the incentives for attention, this is the relevant notion of connectedness that ensures that the emotions of the two messages have to be determined jointly. More specifically, the following condition ensure that the set of messages consistent with a partition cannot be divided into disjoint subsets such that emotions can be chosen independently on each subset.

Assumption 5 (Partitional Feedback Structure). *There is a partition $\mathcal{P}$ of Ω such that*

$$\mathcal{F}(a, \omega) = (v(a, \omega), a, \mathcal{P}(\omega))$$

where $\mathcal{P}(\omega)$ is the partition cell that contains ω . We define Z as the image of $\mathcal{F}$ in $\mathbb{R} \times A \times \mathcal{P}$.

For each partition cell $P \in \mathcal{P}$, the subset of the message space that is consistent with this partition cell, $\{(v, a, P) \in Z | v \in \mathbb{R}, a \in A\}$, is a connected subgraph of Z .

For simplicity, we analyze the model under a known bias $\beta \in (0, 1)$. Thus, we consider a pure hidden-action problem without hidden information.

Assumption 6. *The distribution of $\beta \in (0, 1)$ is degenerate.*

Under known β , we can generalize the model in two dimensions. First, we drop the exchangeability assumption on v and the restriction to mechanisms with exchangeable $v + m$. Second, we generalize the entropy-based attention cost to any posterior-separable cost function. To be coherent with the notation above, we continue to define the attention cost in terms of the state-dependent stochastic choice function.²¹ For each state-dependent stochastic choice function $p: \Omega \rightarrow \Delta(A)$, let $\tau(p) \in \Delta(\Delta(\Omega))$ denote the distribution over posteriors $\gamma \in \Delta(\Omega)$ induced by p via Bayesian updating.

Definition 11 (Posterior-Separable attention cost). *The attention cost c is posterior-separable if*

$$c(p) = \mathbb{E}_{\tau(p)}[H(\mu) - H(\gamma)]$$

where $H: \Delta(\Omega) \rightarrow \mathbb{R}$ is strictly concave and differentiable on $\Delta(\Omega)$.

Posterior-separable attention costs arise naturally from several foundations, including information theory (Sims, 2003), sequential sampling (Morris and Strack, 2019; Hébert and Woodford,

²¹This is without loss if the function H below is strictly concave. As is well-known, if we defined the attention cost as a function of the acquired signal structure, we can without loss identify the signal with the action under strictly concave H .

2023), and constant marginal cost of experimentation (Pomatto, Strack, and Tamuz, 2023). The function H can be understood as a measure of uncertainty of a distribution, with entropy as an example. Thus, under a posterior-separable cost, the agent incurs a cost proportional to the expected reduction of the uncertainty of their belief. That H is strictly concave ensures that more information (in the Blackwell sense) is more costly.

Assumption 7. *The attention cost $c(p)$ is posterior separable.*

Given incomplete feedback and the absence of hidden information, the new principal's problem P^{inc} is to pick the solution with minimal variance of emotions to:

$$\max_{m,p} -c(p) + \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p(a|\omega) v(a, \omega) \quad (P^{\text{inc}})$$

s.t.

$$p \in \arg \max_{\tilde{p}} -c(\tilde{p}) + \beta \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) \tilde{p}(a|\omega) (v(a, \omega) + m(a, \omega)) \quad (\text{IC})$$

$$\sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p(a|\omega) m(a, \omega) = 0 \quad (\text{BC})$$

$$\forall a, a' \in A, \omega, \omega' \in \Omega: \mathcal{F}(a, \omega) = \mathcal{F}(a', \omega') \rightarrow m(a, \omega) = m(a', \omega') \quad (\text{F})$$

We introduce the following class of solutions, which we interpret below as a combination of regret and disappointment. To simplify the exposure, we assume that the first-best attention strategy p , which maximizes the principal's objective, is unique. Further, we define the consideration set $S \subseteq A$ as the set of actions that have positive probability according to the first-best attention strategy p , formally $S = \{a \in A \mid \sum_{\omega \in \Omega} \mu(\omega) p(a|\omega) > 0\}$.

Definition 12 (Linear Regret-Disappointment Solution). *A solution (m, p) to the principal's problem P^{inc} is a linear regret-disappointment solution if for all $a \in S, \omega \in \Omega$:*

$$m(a, \omega) = \left(\frac{1}{\beta} - 1 \right) (v(a, \omega) - v(\mathcal{P}(\omega))) \quad (22)$$

where $v(\mathcal{P}(\omega))$ is the expected payoff conditional on partition element $\mathcal{P}(\omega)$ with respect to the first-best attention strategy p ,

$$v(\mathcal{P}(\omega)) = \frac{1}{\mu(\mathcal{P}(\omega))} \sum_{\omega' \in \mathcal{P}(\omega)} \sum_{a \in S} \mu(\omega') p(a|\omega') v(a, \omega').$$

Such a solution has the natural interpretation as setting for each partition cell a reference point or benchmark $v(\mathcal{P}(\omega))$, which corresponds to the expected payoff the agent should achieve – according to the first-best attention strategy – conditional on that partition cell.²² Note that

²²In the Appendix, we show that for non-quadratic costs of emotions, an analogous solution with a different reference point v is optimal.

in contrast to other models of reference points (Kahneman and Tversky, 1979; Gul, 1991; Kőszegi and Rabin, 2006), the optimal gain-loss utility here is linear.

It is instructive to compare (22) to the regret theory by Gabillon (2020), where regret depends on the difference between the realized payoff and the highest payoff that could have been achieved through some action given the ex-post feedback about the state. As in their proposal, the reference point is updated on feedback about the state.²³ However, in our model, the reference point is not the highest attainable counterfactual payoff but instead the *expected* payoff conditional on ex-post feedback. Because this expected payoff is also affected by the payoffs of the chosen action *had different states realized*, one can interpret the emotional payoff as a combination of regret and disappointment. Next, we make this intuition more clear.

Another, mathematically equivalent, formulation of (22) relates the emotional payoff to regret and disappointment. Let $\gamma: A \rightarrow \Delta(\Omega)$ map each action to the posterior associated with it via Bayesian updating, which we write $\gamma(\omega|a)$. We can define, *conditional on partition cell* $\mathcal{P}(\omega)$, the probability, $p(a|\mathcal{P}(\omega))$, and the expected payoff of, $v(a|\mathcal{P}(\omega))$, of a ,

$$\begin{aligned} p(a|\mathcal{P}(\omega)) &:= \sum_{\omega' \in \mathcal{P}(\omega)} \frac{\mu(\omega')}{\mu(\mathcal{P}(\omega))} p(a|\omega') \\ v(a|\mathcal{P}(\omega)) &:= \sum_{\omega' \in \mathcal{P}(\omega)} \frac{\gamma(\omega|a)}{\gamma(\mathcal{P}(\omega)|a)} v(a, \omega'). \end{aligned}$$

Given this notation, we can rearrange terms and rewrite equation (22) as

$$\begin{aligned} m(a, \omega) &= \sum_{a' \neq a} p(a'|\mathcal{P}(\omega)) \left(\frac{1}{\beta} - 1 \right) (v(a, \omega) - v(a'|\mathcal{P}(\omega))) \\ &\quad + p(a|\mathcal{P}(\omega)) \left(\frac{1}{\beta} - 1 \right) (v(a, \omega) - v(a|\mathcal{P}(\omega))). \end{aligned} \tag{23}$$

The term in the first line can be interpreted as regret and rejoicing. The emotion depends on the weighted difference between the actual payoff and the expected payoffs of all other actions a' , conditional on partition cell $\mathcal{P}(\omega)$. The expected payoff of each action is computed with respect to the first-best attention strategy. The regret-like comparisons to different actions are weighted by the probabilities of the respective actions, again conditional on the partition element $\mathcal{P}(\omega)$.

This is in contrast to a widely used theory of regret where the comparison is with respect to the ex-post optimal action (Quiggin, 1994; Sarver, 2008), which allows for regret only and no rejoicing. This proposal is arguably extreme when the agent has a large choice set and does not seem psychologically plausible when the ex-post optimal action was not even considered. Moreover, in some relevant applications as in finance, under no short-sale constraints the ex-post optimal return is infinite (Qin, 2020). In contrast, our prediction corresponds to the proposal of Loomes and

²³To be precise, in (22), the reference point is updated only on feedback about the state that is action-independent, namely what is learned through the partition of the state space. It is not updated on what is learned from the realized payoff $v(a, \omega)$.

Sugden (1982) to generalize regret theory to multiple actions based on such a weighted sum of comparisons to *all* other actions, not just the ex-post optimal one. They highlight the importance of providing a theory of the appropriate action weights. Sugden (1986) writes that

“Any plausible theory of such weights must, I think, take some account of the extent to which an action is a serious candidate for choice or, to put it slightly differently, of the extent to which the individual could sensibly blame himself for not having chosen it.”

In our contexts, the action weights are naturally provided by the state-dependent choice probabilities under the first-best attention strategy. These choice probabilities capture with what probability the agent should have taken some action under the realized state. This implies that regret and rejoicing do not depend on actions outside of the consideration set, which are taken with probability 0. This captures that as Sugden (1986) writes, only actions that are “serious candidates” or “real options” should be sources of regret and rejoicing. While regret and rejoicing are linear here and thus predict no violations of expected utility axioms from choice, linear regret still has effects on menu choice (Sarver, 2008).

The term in the second line of (23) can be interpreted as disappointment and elation. The emotion depends on the difference between the payoff $v(a, \omega)$ and the expected payoff of the chosen action. The expected payoff of the chosen action is computed with respect to the posterior that the agent holds when taking the action. However, in contrast to standard models of disappointment and elation (Bell, 1985; Loomes and Sugden, 1986; Gul, 1991), the expected payoff is conditioned on what is learned about the state. The weight on disappointment and elation is the probability of taking the actual action conditional on the partition cell.

Theorem 3. *Under Assumptions 5 to 6, any solution of P^{inc} is a linear regret-disappointment solution and induces the first-best attention strategy.*

Because we assumed that the bias is known, the optimal mechanism can perfectly offset it by scaling up the utility differences by $1/\beta$. Using a result in Whitmeyer and Zhang (2022), which builds on the Lagrangian Lemma of Caplin, Dean, and Leahy (2022), we show that such inflating of the utility differences is also *necessary*, in order to implement the first-best attention strategy. The remaining goal of the principal is to reduce the variance of emotions, while scaling up the utility differences. This goal is constrained by the feedback structure. Under complete feedback, the second line of (23) is 0, and only regret is used. However, for incomplete feedback structures, both lines of (23) are generally non-zero and accordingly the optimal emotions can be interpreted as a combination of regret and disappointment.

Disappointment and elation are used because, under incomplete feedback, the choice of all emotional utilities consistent with some partition element of states is interrelated (Appendix, Lemma 10). Thus, all payoffs under that partition element, including the payoffs of the chosen action *had another state realized*, affect the optimal emotions under some feedback signal.²⁴

²⁴This statement holds also under unknown bias.

Thus, in our model, disappointment and elation can be thought of as *second-best* motivators for attention when complete feedback is not available. They are second-best because they lead to a higher variance of emotions than using regret and rejoicing.²⁵

One might be surprised that disappointment and elation motivate more attention because they depend on the realization of the state, which the agent cannot affect. By contrast, regret and rejoicing seem to depend more directly on the agent's action, which perhaps more intuitively incentivizes attention. However, although the agent cannot choose the unconditional distribution of states, he can implicitly choose the distribution of states *conditional* on an action by paying attention. Indeed, this is the perspective of attention taken by the posterior approach, in which the agent acquires posteriors over states conditional on each taken action (Caplin and Dean, 2013). This goes to say that, like regret and rejoicing, the emotions of disappointment and elation penalize bad outcomes and reward good outcomes and thus raise the return to attention. As argued above, regret and rejoicing raise the return to attention in the optimal way but under constraints through the feedback structure, disappointment and elation have a role to play.

B Proof of Theorem 3

Because the feedback structure reveals the material payoff by Assumption 5, the principal can choose the utility $u(z)$ for each message $z = (v, a, P) \in Z$. For $z = (v, a, P)$, we write $v(z)$ and $a(z)$ for the first and third component, respectively. The following Lemma uses of the fact that the optimal attention strategy under posterior-separable attention costs depends on the state-wise utility differences as shown in Whitmeyer and Zhang (2022).

Lemma 10. *A first-best attention strategy is induced only if for all elements z_1, z_2 of Z that are directly connected by an edge and have $a(z_1), a(z_2) \in S$,*

$$u(z_1) - u(z_2) = \frac{1}{\beta}(v(z_1) - v(z_2)).$$

Proof. By Theorem 4.3 in Whitmeyer and Zhang (2022), to induce an attention strategy that is optimal for the principal, the relative incentives, such as $u(a, \omega) - u(a_1, \omega)$ for $a \in S, \omega \in \Omega$, are uniquely pinned down. Since the agent discounts the relative incentives by β , for the relative incentives to equal those of the principal, they need to be scaled up by $\frac{1}{\beta}$. This is achieved if and only if for all $\omega \in \Omega$, $u(a, \omega) - u(a_1, \omega) = \frac{1}{\beta}(v(a, \omega) - v(a_1, \omega))$. This in turn is equivalent to all elements m_1, m_2 of Z that are directly connected by an edge (that is, are achieved at the same state) to satisfy $u(m_1) - u(m_2) = \frac{1}{\beta}(v(m_1) - v(m_2))$. $\square$

The above is an only-if statement. It ensures only that if we restricted the agent's choice to the consideration set S , a first-best attention strategy would be incentive-compatible (Whitmeyer and Zhang, 2022). To implement the first-best attention strategy when the whole decision problem A

²⁵Under an unknown bias, as we have shown under complete feedback, regret and rejoicing are not only optimal because they minimize the variance of emotions but also because they allow tailoring the incentives to the state.

is feasible, it is sufficient to punish the actions in $A \setminus S$ with sufficiently negative emotions. As long as the principal deters the agent from choosing any action in $A \setminus S$, she is indifferent with regards to how strongly she punishes these actions as it does not affect her utility nor the implementation cost of realized emotions.

The utility difference being scaled up by $\frac{1}{\beta}$ is a transitive relation that extends to all nodes on the largest connected subgraph. If z and $\tilde{z}$ are on the same connected subgraph of Z , then there is a finite chain of elements $(z_1, \dots, z_k)$ of Z such that $z = z_1$, $\tilde{z} = z_k$, and each adjacent pair is directly connected. If each of those directly connected pairs has a utility difference scaled up by $\frac{1}{\beta}$, then

$$\begin{aligned} u(\tilde{z}) - u(z) &= (u(z_k) - u(z_{k-1})) + \dots + (u(z_2) - u(z_1)) \\ &= \frac{1}{\beta}(v(z_k) - v(z_{k-1})) + \dots + \frac{1}{\beta}(v(z_2) - v(z_1)) = \frac{1}{\beta}(v(\tilde{z}) - v(z)). \end{aligned}$$

Conversely, if the utility difference of all pairs of nodes on the same connected subgraph is scaled up by $1/\beta$, then trivially the utility difference of directly connected nodes is scaled up by $1/\beta$.

Thus, an optimal attention strategy of the principal is implemented only if the utility difference of all elements on the same connected subgraph are scaled up by $1/\beta$. By Assumption 5, the sets of all messages that are consistent with some partition element $P \in \mathcal{P}$ are connected. By the definition of edges, such sets for different partition elements are not directly connected. In other words, Z has one connected subgraph for each partition element $P \in \mathcal{P}$. This pins down the utility function u except for one constant per subgraph, which is determined by the implementation-cost minimization problem. More specifically, by the above, we can write

$$m(a, \omega) = \left(\frac{1}{\beta} - 1 \right) (v(a, \omega) - v(\mathcal{P}(\omega))). \quad (24)$$

The remaining problem is to determine the “reference point” $v(\mathcal{P}(\omega))$ for each partition cell.

The optimal reference point is connected to the concept of central tendency from statistics. The central tendency $\mathcal{C}_f(X)$ of random variable X with respect to a strictly convex function f is defined as

$$\mathcal{C}_f(X) := \arg \min_c \mathbb{E}[f(X - c)],$$

which is known to be unique under strict convexity of f . Here, we consider the central tendency $\mathcal{C}_\psi$ with respect to the strictly convex implementation cost ψ .

Lemma 11. *Under implementation cost $\psi(x) = |x|^k$ with $k > 1$, the implementation cost minimizing solution has reference point $v(P)$ equal to the central tendency of payoff v on the partition element P with respect to the optimal attention strategy.*

Proof. First, the centrality measure of emotions must be 0, otherwise we could decrease the implementation cost. Let $P \in \mathcal{P}$. The attention strategy that is implemented defines a distribution over emotions. The implementation cost on this partition is the expectation of $\psi(u(a, \omega) - v(a, \omega))$.

Under the optimal emotions, $\psi(u(a, \omega) - v(a, \omega) - c)$ is minimized at $c = 0$, otherwise one could further reduce the implementation cost. This implies by definition that the central tendency of emotions is 0.

Next, we show two properties of the central tendency. First, $\mathcal{C}_\psi(X + b) = \mathcal{C}_\psi(X) + b$ due to

$$\arg \min_c \mathbb{E}[\psi(X + b - c)] = b + \arg \min_c \mathbb{E}[\psi(X - c)].$$

Second, under $\psi(x) = |x|^k$ with $k > 1$, $\mathcal{C}_\psi(aX) = a\mathcal{C}_\psi(X)$ for all $a \in \mathbb{R}$. First of all, note that $\psi(ax) = |ax|^k = (|a||x|)^k = |a|^k|x|^k = |a|^k\psi(x)$. Then,

$$\begin{aligned} \arg \min_c \mathbb{E}[\psi(aX - c)] &= a \arg \min_c \mathbb{E}[\psi(a(X - c))] = a \arg \min_c \mathbb{E}[|a|^k \psi(X - c)] = \\ &= a \arg \min_c \left(|a|^k \mathbb{E}[\psi(X - c)] \right) = a \arg \min_c \mathbb{E}[\psi(X - c)]. \end{aligned}$$

Finally, using the two previous facts we show that the reference point of P is the central tendency of the payoff v , with respect to the distribution induced by the implemented attention strategy on P . The reference point $v(\mathcal{P}(\omega))$ is the (hypothetical) payoff $\tilde{v}$ that obtains emotions 0 according to equation (24). By equation (24), emotions are a linear function of the payoff $v(a, \omega)$. Then, the realized payoff is also a linear function f of emotions. The payoff $\tilde{v}$ is thus the image of 0 under f , $\tilde{v} = f(0)$. By the above, $\mathcal{C}_\psi(aX + b) = a\mathcal{C}_\psi(X) + b$, so $\mathcal{C}_\psi$ commutes with linear functions. By 0 being the central tendency of emotions, $\tilde{v}$ is the central tendency of the payoff. $\square$

As is known, the central tendency $\mathcal{C}_\psi(X)$ of random variable X with respect to $\psi(x) = x^2$ equals the expectation of X . This concludes the proof.

VII Appendix C: Interpretation as Contracting with an Expert

As mentioned in the main text, the model can be interpreted in a more conventional way as a principal hiring an agent (expert) to acquire information. Of particular relevance are CEO and fund manager compensation.

The principal reaps utility $v(a, \omega)$ from action a under state ω and pays monetary transfers (e.g., wages) $m(a, \omega)$ to the agent, which are valued quasi-linearly by the agent. The bias of the expert can capture both a conflict of interest between principal and agent, as well as a behavioral bias of the expert. We discuss both cases in turn.

A Expert with Conflicting Interests

Our model can be interpreted as the optimal monetary contract between a principal and an expert with a conflict of interest over how much information to acquire. In particular, assume that both principal and agent care about the information cost as well as the decision outcome but differ in how they weigh these considerations. For instance, investors or shareholders may internalize the

time cost of information gathering to some degree due to the resulting decision delay or because it diverts the agent from other valuable activities. CEOs or fund managers may care about the decision outcome out of intrinsic motivation or reputational concerns. However, from the principal's perspective, the agent might overemphasize the information cost in the absence of financial incentives, potentially leading to suboptimal information acquisition. In these contexts, it is realistic to assume that information acquisition is unverifiable and only the outcome from decision-making is contractible. Information acquisition often does not produce tangible evidence and even when it does, the evidence might be hard to communicate or verify without the necessary expertise on the principal's side (Lindbeck and Weibull, 2020). Finally, we take serious the possibility that the principal is uncertain about how much the agent overweights the information cost, by incorporating hidden information.²⁶

The last model component to reinterpret is the cost of emotions ψ . Under a quadratic cost $\psi(x) = x^2$, our cost-of-emotions refinement is equivalent to an *ex-ante budget constraint* plus the refinement of reducing the variance of transfers. Note that the mean squared transfers equal the squared mean transfer plus the variance of transfers, as for any random variable X , $\mathbb{E}[X^2] = \mathbb{E}[X]^2 + \text{Var}[X]$. Thus, minimizing the quadratic cost is equivalent to minimizing the sum of the squared expected transfers and the variance of transfers. The principal can add a constant to transfers uniformly to reduce the expected transfers to 0, without affecting the variance of transfers. Further, this does not affect the agent's behavior. Thus, minimizing the quadratic cost ψ is equivalent to an ex-ante budget constraint

$$\int \left(\sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p_{\beta}(a|\omega) m(a, \omega) \right) dF(\beta) = 0 \quad (\text{BC})$$

and the refinement of minimizing the variance of transfers,

$$\int \left(\sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p_{\beta}(a|\omega) m(a, \omega)^2 \right) dF(\beta).$$

Under an interpersonal interpretation with m as monetary transfers, Ψ can be thought of as a implementation cost or as a cost of deviating from an ex-ante budget. Further, when interpreting the principal's problem in organizational contexts where monetary transfers are not used, we can interpret transfers as the reduced-form utility of reward- and punishment-actions taken by the principal. Rewards, such as praise and gifts, as well as punishments, such as making the agent perform unfavorable tasks (which require monitoring), might both be costly for the principal, as captured by ψ .²⁷

²⁶A shortcoming of this application is that our model does not feature a limited liability constraint, which binds if the budget b is low enough.

²⁷See Andreoni, Harbaugh, and Vesterlund (2003) for more examples of rewards and punishments in different social and institutional contexts.

B Dynamically-Inconsistent Expert

If we interpret bias as a behavioral bias of the agent, then our optimal contract can be readily interpreted as the *welfare-maximizing* commitment contract for the agent, subject to an ex-ante *budget balance constraint*, if we again assume a quadratic cost $\psi(x) = x^2$.²⁸

Furthermore, as demonstrated below, our model also applies to a *profit-maximizing* principal who hires a dynamically-inconsistent expert, subject to a *participation constraint*. Here, the principal values the decision outcome net of transfers paid to the agent, but does not care about the agent’s attention cost directly. The agent, on the other hand, cares only about the transfers received net of the attention cost. Instead of a budget constraint, the principal faces an ex-ante participation constraint that guarantees the agent a certain expected utility before his bias realizes. The participation constraint effectively makes the principal internalize the agent’s attention cost, rendering the problem equivalent to ours.

In these contexts, our work contribute to the understanding of compensation schemes, specifically those that assess the agent’s performance relative to a peer benchmark. Relative performance evaluation is common in compensation schemes for fund managers (Ma, Tang, and Gómez, 2019; Evans, Gomez, Ma, and Tang, 2023) and, to some degree, for CEOs (Edmans, Gabaix, and Jenter, 2017).²⁹ Such benchmarks can be justified by by Holmström’s (1979) informativeness criterion in standard moral-hazard settings, because subtracting exogenous common shocks to the return of the industry increases the informativeness of the performance evaluation. Our model offers a different explanation based on incentives for information acquisition. Peer comparisons can be interpreted as counterfactual comparisons, since payoffs of relevant peers inform the principal about the payoff that alternative actions would have lead to. Indeed, Theorem 3 shows that under a known bias, transfers proportional to the difference between the realized return and the expected return under the realized market conditions are optimal.

Consider a principal who hires a dynamically-inconsistent agent (expert) to acquire information.³⁰ The principal reaps utility $v(a, \omega)$ from action a under state ω and pays monetary transfers (wages) $m(a, \omega)$ to the agent. The agent maximizes transfers paid to them. Both have quasi-linear utility. After signing the contract, the agent faces a stochastic temptation shock to his preferences, as in random-indulgence models. Accordingly, the individual-rationality constraint needs to hold

²⁸At first glance, a welfare-maximizing principal would have a different objective than our principal because she would care about the agent’s transfers. However, this is immaterial as transfers cancel out in expectation due to the ex-ante budget constraint.

²⁹Evans, Gomez, Ma, and Tang (2023) find that 71% of mutual funds in the US use peer-benchmarked compensation.

³⁰See Kaur, Kremer, and Mullainathan (2015) for evidence on self-control issues at work.

in expectation. The principal's problem is

$$\begin{aligned}
& \max_{m, (p_\beta)_\beta} \int \left(-c(p_\beta) + \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p_\beta(a|\omega) (v(a, \omega) - m(a, \omega)) \right) dF(\beta) \\
& \quad \text{s.t.} \\
& \quad \forall \beta: p_\beta \in \arg \max_p -c(p) + \beta \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p(a|\omega) m(a, \omega) \quad (\text{IC}) \\
& \quad \int \left(-c(p_\beta) + \beta \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p_\beta(a|\omega) m(a, \omega) \right) dF(\beta) \geq U_0 \quad (\text{IR})
\end{aligned}$$

If the agent was dynamically consistent ($\beta \equiv 1$), the principal would achieve the first-best by selling the firm to the agent. In the presence of a bias, the principal can be seen as maintaining control of the firm in order to provide a self-control mechanism to the agent for their mutual benefit.

By standard arguments, the (IR) constraint is binding. Inserting the (IR) constraint into the objective, we obtain the following equivalent formulation.

$$\begin{aligned}
& \max_{m, (p_\beta)_\beta} \int \left(-c(p_\beta) + \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p_\beta(a|\omega) v(a, \omega) \right) dF(\beta) - U_0 \\
& \quad \text{s.t.} \\
& \quad \forall \beta: p_\beta \in \arg \max_p -c(p) + \beta \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p(a|\omega) m(a, \omega) \quad (\text{IC}) \\
& \quad \int \left(-c(p_\beta) + \beta \sum_{\omega \in \Omega} \sum_{a \in A} \mu(\omega) p_\beta(a|\omega) m(a, \omega) \right) dF(\beta) = U_0 \quad (\text{IR}')
\end{aligned}$$

The constant $-U_0$ in the principal's objective can be ignored. Identifying m with $v + m$ in the main text, the objective and (IC) constraint are identical. The (IR') constraint can be satisfied by adding a uniform constant to emotions, so (IR') acts in effect as an ex-ante budget constraint. Adding the refinement that the principal chooses the mechanism with lowest variance of transfers, renders the problem essentially equivalent to the principal's problem in the main text.³¹ The only difference is that the optimal transfers emulate the payoff v in addition to the emotions in the main formulation. That is because in this formulation the agent has no direct interest in the decision. Additionally, the optimal transfers are uniformly shifted by a constant, which depends on the outside option U_0 .

³¹The equivalence also holds under an appropriate exchangeability constraint or when the principal chooses a menu of transfer schemes.